

Heredity can precede individuality in a digital primordial soup

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Review copy, 2026-10-09 (revision 2). Nature Article format: summary ≤ 200 words, main text $\approx 3,600$ words, six figures, Methods, Extended Data legends and a Supplementary Information with the theorems and proofs. Every number is from a generated table of the project repository; every experiment reported is complete.

Theories of the origin of life disagree about what came first but agree on the transition: a spreading pattern became an individual whose future depends on itself, not on its surroundings¹⁻⁴. We watched, repeated and dissected it in thousands of worlds of random machine code. On tapes of 16 to 64 bytes with harmless by-products, in the Z80 and an 8080-like subset, the first replicator is a two-byte word that writes itself when executed, runs on into its neighbour and makes damaged copies that vary with the neighbour; its lineages soon stop copying. Closure, a return of control flow into the organism, follows where a closing design is within reach, and seeded closers displace the open form. Closed copies no longer depend on the partner, but closers differ in heredity, and in the commonest closing core one byte decides how: the byte that sets where the copy lands sets how many sites a lineage can transmit. Closers that rewrite themselves from their first bytes erase mutations; closers that copy the whole tape keep them. The environment chooses: regenerators prevail where by-products are harmless, transmitters where they are lethal. In a second machine one instruction decides whether life begins open.

Main

What distinguishes a living thing from the chemistry around it has two answers: a list of properties, metabolism, compartment and heredity, with the question of which came first⁵⁻⁹, or a relation: a living system is one whose operations produce the organisation that produces them, so that its future depends on its own past rather than on its environment, in Maturana and Varela's operational closure¹, Rosen's closure to efficient causation², Fontana and Buss's organisational closure^{3,10} and, made quantitative, the information-theoretic individuality of Krakauer and colleagues^{4,11}. This is also the moment at which population genetics places the start of Darwinian evolution proper, when vertical descent comes to dominate communal exchange^{12,13}. No experimental system has allowed this transition to be observed as it happens, repeated under controlled variation, and taken apart.

Digital organisms usually descend from a designed ancestor^{14,15}; random-program soups come closer. When random byte strings for a simple machine execute one another, with no fitness function and no replacement, self-replicating programs arise and take over¹⁶, as they did in the Coreworld^{17,18}, among random Lisp expressions¹⁹ and in artificial chemistries^{20,21}. For the Z80 processor the first replicators are known to be load-push programs that give way to a block-copy family^{16,22}, a takeover attributed to mutational robustness²². We treated such a soup as a test of one hypothesis: that the smallest programs that write themselves run on into their surroundings, so that life begins open, and that individuality, a path of execution that returns into the organism, is a later invention. We ran more than 3,600 worlds of random Z80 code, varying the instruction set, program length, execution budget, mutation rate and memory geometry, with confirmatory stages pre-registered with kill criteria (Methods). Heredity is measured by intervention, as in serial-transfer tests of self-replication²³: a program is lifted out and cultured against fresh random partners, so that a pattern that merely spreads is not mistaken for one that inherits. We then repeated the central experiment on the Z80 restricted to an 8080-like subset of its instructions, and on a second, minimal machine whose instruction set we could edit. Because the most common tape can misrepresent a population, we classified random cells and followed single mutations through serial transfers, and found that closure comes in two kinds, which one byte of the genome sets and the environment chooses between.

Order before life

The soup holds 20,000 programs of L bytes on a 160×125 lattice (Fig. 1a). At each step 8,192 random cells are each paired with a lattice neighbour, the pair is executed as one program for 128 instructions and written back, and a background mutation replaces a fixed number of random bytes; nothing else happens (Methods). A tape is *heritable* if, lifted out and executed against random partners, it makes copies that copy in turn; emergence is dated by the first heritable tape among the most common, and the heritable fraction of random cells is the population measure.

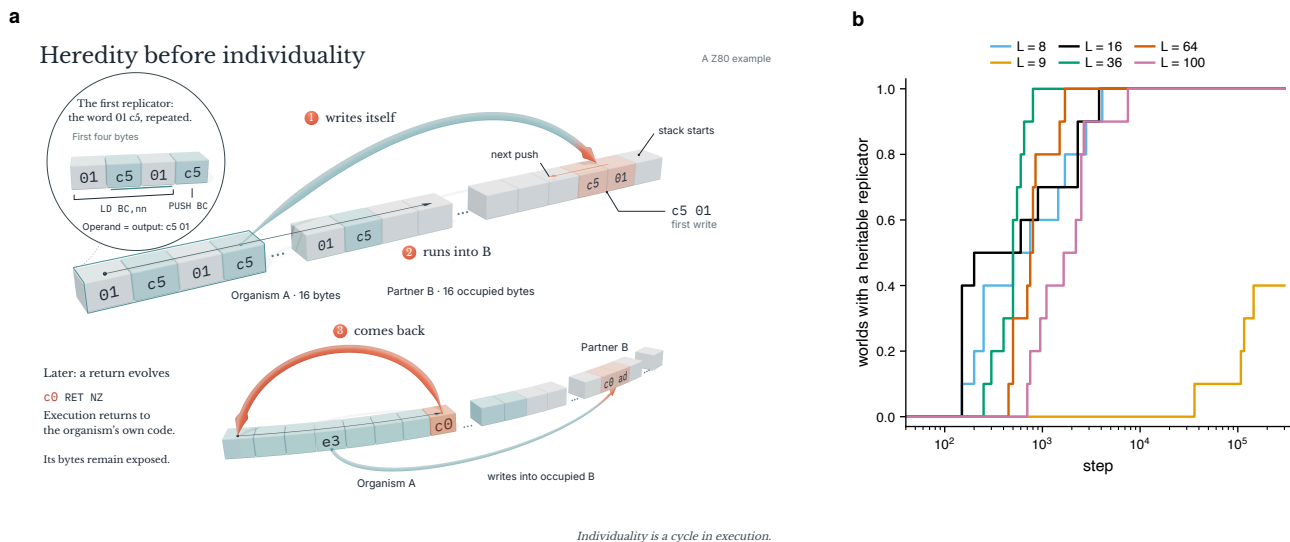


Fig. 1 | The first replicator and its closure. **a**, The first replicator and its encounter. Organism A (teal, 16 bytes) and partner B (grey, 16 bytes) share one 32-byte memory; execution starts at A's first byte and moves right, the stack pointer starts at B's last byte and moves two bytes left with every push. The first replicator is the word 01 c5 repeated: LD BC, nn loads its own next two bytes, c5 01, and PUSH BC writes exactly those two bytes at the stack pointer (operand = output), so the partner fills from its far end with the organism's own word (1) while execution runs on into the partner (2). Later a return evolves (3): execution revisits the organism's own code, and its bytes remain as exposed as before. **b**, Kaplan–Meier fraction of worlds with a heritable replicator by step for L = 8–100 (Stage E, 10 worlds per length, 300,000 steps).

The first thing to form is order that does not inherit. When the most common pattern first occupies a tenth of the soup, 24–34% of all bytes are zero at every tape length from 9 to 100 (Stage B, Methods; one world in Fig. 2). The zeros are self-written: push, call and reset instructions executed with empty registers write zeros wherever the stack pointer points, and zero is itself an instruction that does nothing, so the pattern spreads by being written and changes nothing when executed. We call this flood *tar*, after the asphalt that prebiotic chemistry makes from the reactions that also make sugars²⁴. The tar is what detectors see: the high-order entropy used to detect the transition in random-program soups¹⁶ is excellent at L = 16 but near chance or inverted in unablated worlds at six of eight lengths from 25 to 100 bytes, a complexity-times-abundance measure of selection^{25,26} passes ten times its random-soup value before the first heritable replicator in 71 of 80 worlds, and the first replicator's assembly index is the minimum for its length, shared by every tape that repeats a two-byte word, tar included (Extended Data Fig. 1). Order comes first, it is as simple as what follows, and only intervention tells the two apart.

The first replicator is open

The first heritable replicator is the same program everywhere: the two-byte word 01 c5, or one of its register variants, repeated to fill the tape (Fig. 1a). On tapes of 16 to 64 bytes with benign tar it comes first in 170 of 170 worlds, across five lengths, the Z80 and its 8080-like subset, and two pairing rules, at Kaplan–Meier medians of 300–750 steps (100 with well-mixed pairing; Methods); on 81- and 100-byte tapes it is also a load–push word in 20 of 20 and 18 of 20 worlds, though these were not traced, and on shorter tapes block copiers come first (Fig. 1b). The order does not rest on the harness's conventions: with every register drawn at random before each encounter the first replicator is a load–push word in 20 of 20 worlds, and with a random stack pointer it is open in 18 of 20 (Methods). Executed, the word loads its own next two bytes into a register pair and writes that pair where the stack pointer points, in the partner: the program is its own data, and it copies itself as a side effect of being run. It has no loop, no counter and no test of its surroundings. It is also nearly the only thing of its size that replicates: of all 65,536 two-byte words tiled to 16 bytes, exactly five, besides the trivial zero word, copy themselves into an all-zero partner, all of them load-and-push pairs (Extended Data Fig. 2; Supplementary Information, Proposition 3). Removing the instructions it uses removes it: without immediate loads 2 of 10 worlds produce a replicator by 300,000 steps, and without the stack writers life arrives 160- to 992-fold later as a block-copy replicator; removing any other family changed the median time to the first replicator by 0.75- to 6.5-fold, with no seed-paired sign test below P = 0.18 (Extended Data Fig. 3).

The first replicator is *open*. Its pointer fetches partner bytes in every one of 256 encounters, for every traced first replicator

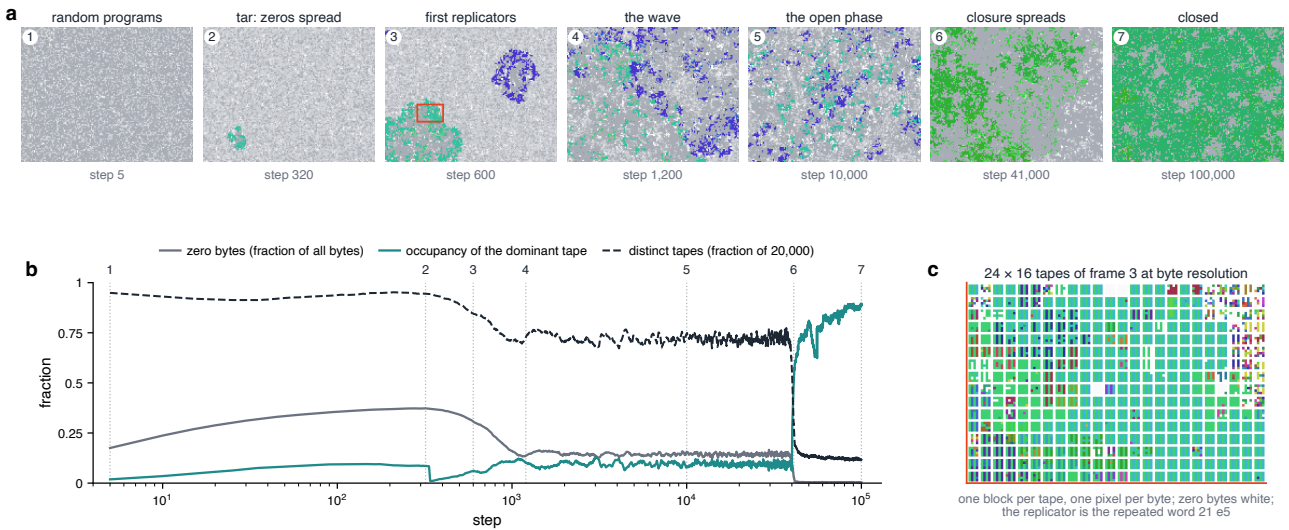


Fig. 2 | One world, watched. Seed 2002 at $L = 16$, run with dense snapshots for Supplementary Video 1. **a**, The lattice (160×125 tapes, one pixel per tape) at seven steps. The four most common tapes with at least 20 copies are coloured by the mean colour of their bytes (the colour code of c and of the video); every other tape is grey, lighter the more of its bytes are zero, so the tar shows as bleaching. Zeros spread first (frame 2: 37% of all bytes at step 320); the first replicators appear as patches of their own copies (3; the teal patch is the word 21 e5, the blue one 01 c5, both load–push words); they sweep the lattice (4) and compete for 40,000 steps among pockets of zeros (5); a new class carrying a conditional return (c0, RET NZ) spreads from step 41,000 (6) and takes over while the zeros disappear (7): at step 100,000 the dominant tape occupies 89% of the soup, 2,290 distinct tapes remain of 20,000, and 0.4% of bytes are zero. **b**, The same world’s zero-byte fraction, dominant-tape occupancy (cells within Hamming distance 4 of the most common tape) and number of distinct tapes against step; dotted lines, the frames of a. **c**, The 24×16 tapes inside the red rectangle of frame 3 at byte resolution (one 4×4 block per tape, one pixel per byte, zero bytes white): the replicator’s tapes are the repeated word; the rest is random code and tar.

under benign tar (100 of 100 traced, Stages G and K; Methods). Against 256 random partners it copies into a median 0.66–0.73 of them and loses a quarter or more of its own bytes in a median 0.06–0.43 of encounters (Fig. 3a,b), because after writing its copy it executes the partner’s bytes, which can poison the register pair before the next push. With the organism fixed and execution deterministic, the information that flows from the partner into the offspring equals the entropy of the offspring over random partners (Methods). The offspring is an exact copy of the parent in a median 58% of encounters at $L = 16$, 32% at $L = 20$, 5% at $L = 50$ and almost never at $L = 64$; every one of its L positions varies across partners; and its entropy over 256 partners runs from 3.9 bits at $L = 16$ to the 8-bit ceiling of the sample at $L = 64$ (Fig. 3c): what a child looks like depends in part on the neighbour it was made against. In the population the median heritable fraction of random cells stays between 0.09 and 0.19 from step 2,000 to step 20,000 at $L = 16$ (Fig. 3d): the first replicator’s reign is a world of damaged copies. Its lineages do not last either: copied serially into fresh random partners, 96–100% of them stop copying within eight transfers (Fig. 4a).

Closure, where a closer is within reach

The first replicator does not stay. At $L = 16$ the dominant tape carries a control-flow instruction in 20 of 20 worlds by 300,000 steps, 17 of them ending on the identical eight-byte sequence whose conditional return sends the pointer back into the organism’s own body (Fig. 3e; Extended Data Table 1). Every such successor copies into all 256 random partners without self-damage (Fig. 3a,b,f), and in 63 of 67 loop-bearing final dominants the offspring is one and the same string in all 256 encounters (Fig. 3c). Populations agree: at the last snapshot every sampled heritable cell is confined in all 20 worlds at $L = 16$, and most are in 19 of 20 worlds at $L = 20$, 12 of 20 at $L = 32$ and 8 of 20 at $L = 64$ (Fig. 4d), and the heritable fraction of random cells rises to 0.75–0.94 at $L = 16$ (Fig. 3d).

We call this property *closure*. A loop in the code, confinement of the pointer to the organism’s cells and independence of the offspring from the partner travel together here, and all three were measured: every loop-bearing final dominant fetches no partner byte as instruction in any traced encounter (89 of 89), and confinement and zero inflow agree in 106 of 110 finals, the four exceptions being confined finals that keep one to three partner bytes (Extended Data Fig. 4). Closure in this sense is operational closure¹ made countable and one component of individuality⁴ made measurable by intervention. It is not a membrane, since the organism’s bytes are as exposed as before, nor a property of place²⁷: well-mixed pairing delayed it (7 of 10 worlds closed against 20 of 20, Fisher $P = 0.03$; Methods). The return and the block copy work at every

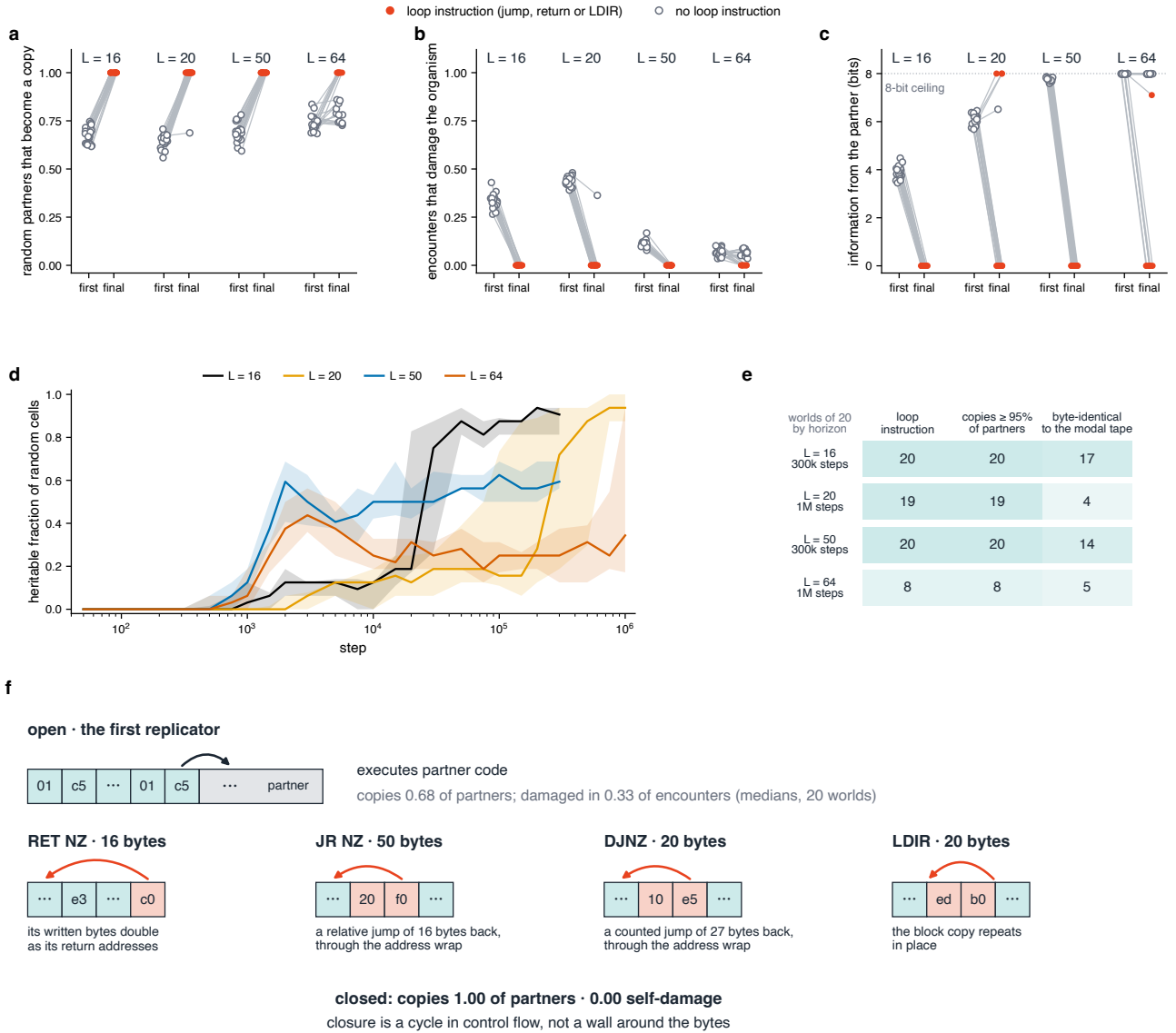


Fig. 3 | The first replicator is open, the successor closed. **a**, Fraction of 256 random partners that become a $\geq 75\%$ copy of the first heritable replicator (first) and of the final dominant (final) of every Stage G world, grouped by tape length (20 worlds per L; grey lines join the same world; open circles, no loop instruction; filled, jump, return or LDIR). **b**, Fraction of encounters in which the organism loses $\geq 25\%$ of its bytes. **c**, Information the offspring carries about the partner, $H(o | x)$ in bits over 256 random partners (dotted line, the 8-bit ceiling of 256 partners). **d**, Heritable fraction of 16 random cells against step (median and interquartile range over worlds). **e**, Worlds (of 20) whose final dominant carries a loop instruction, copies $\geq 95\%$ of partners, and equals the modal final tape byte for byte, by tape length and horizon; at L = 50 the final dominant is confined but its offspring is not (Methods). **f**, Control flow of the first replicator and of four closed successors, schematically (bytes abbreviated; the instruction that closes the cycle is shaded and its return drawn as a vermillion arc): the pusher's pointer runs on into the partner, whereas RET NZ returns to addresses formed from the organism's own written bytes, JR NZ and DJNZ jump back (through the 16-bit address wrap), and LDIR repeats in place. Medians of the partner test over the Stage G worlds at L = 16: the first replicator copies 0.68 of partners and is damaged in 0.33 of encounters; every closed successor copies 1.00 with 0.00 self-damage. Exact traces, Extended Data Fig. 4.

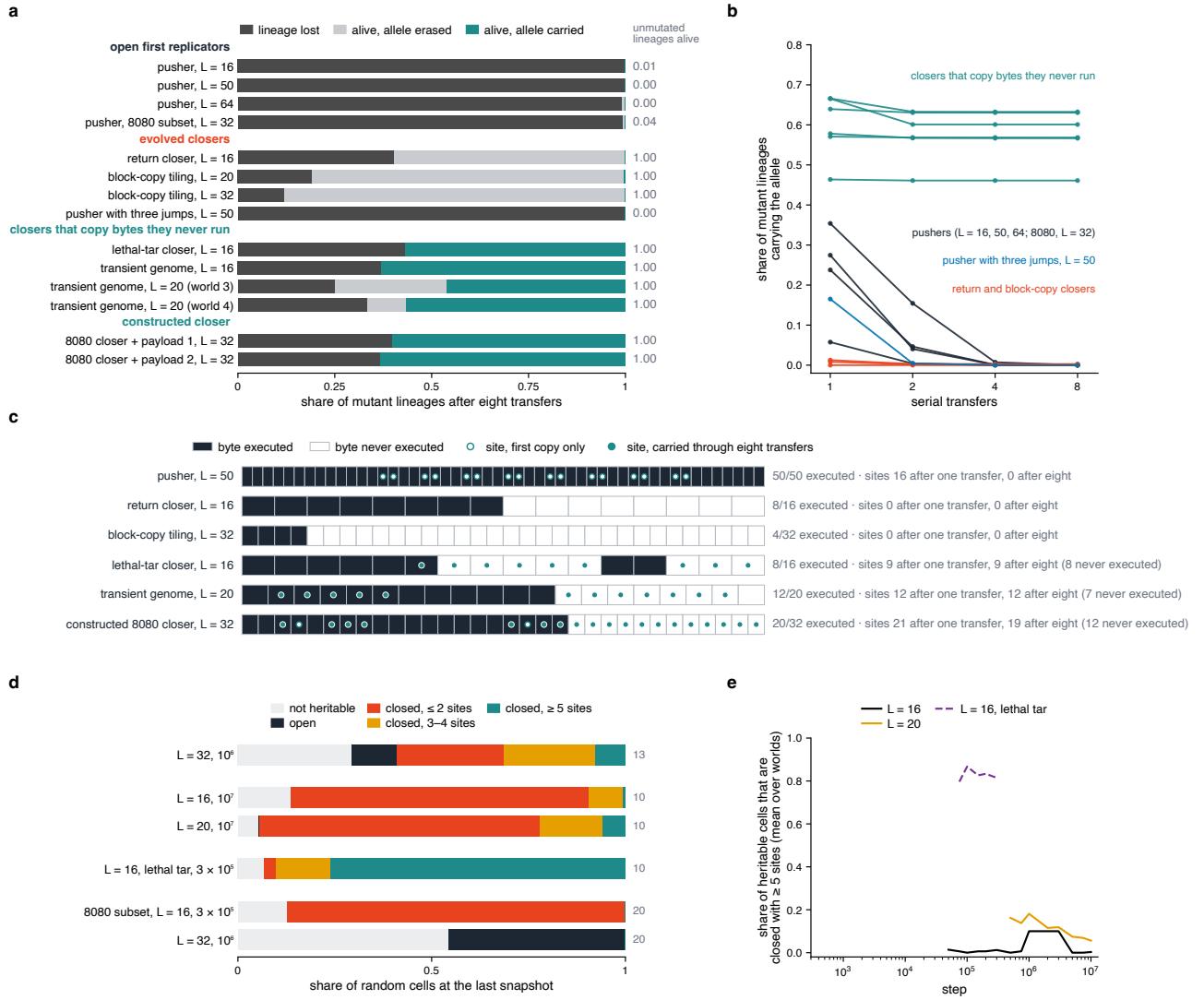


Fig. 4 | Closure regenerates or transmits. a, Fate of every single-byte mutant of 14 genomes after eight serial transfers into fresh random partners (16 lineages per mutant; all 255 alternative bytes per position at $L \leq 32$, 32 seeded values at $L = 50$ and 64): the lineage was lost (a transfer failed to make a $\geq 75\%$ copy), is alive with the mutant byte erased, or is alive carrying it (the mutant's five-byte window anywhere in the tape); every lineage is counted. Right, the share of 256 unmutated lineages still copying after eight transfers. **b**, Share of all mutant lineages carrying the allele against the number of transfers, one line per genome (colours as in a; blue, the $L = 50$ tape with three jumps). **c**, Six of the genomes byte by byte: dark, fetched as instruction stream (opcode or operand) by the unmutated tape in any of 64 encounters; white, never fetched; open circle, a site whose alleles reach first-generation copies only; filled circle, a site whose alleles are still carried after eight transfers (for at least half of its values, the share of live lineages carrying the allele exceeds the chance background by at least 0.5). **d**, Classes of 64 random cells at the last snapshot, mean over worlds (number of worlds at right): not heritable; heritable and open (the pointer fetches a partner byte in at least one of 16 encounters); closed, by the number of transmissible sites in a single-mutant scan of up to 16 heritable cells per snapshot (at most two: regenerators; five or more: transmitters). **e**, Share of heritable cells that are closed with five or more sites against step, mean over worlds, at the snapshots at which at least 80% of a series' worlds are recorded and at least a fifth of the cells are heritable. Sources: results/serial_retention/, results/population/.

length, though both rely on registers that the harness empties before each encounter (below); the relative jump relies on the harness's 16-bit program counter in a memory addressed modulo $2L$, which lands a backward jump inside the organism only at $L = 20$ and 50 , and 25 of the 67 closers carry such a jump (Methods). At the aligned lengths closure is slower and rarer the longer the tape (trend test $P = 6 \times 10^{-5}$): 20 of 20 worlds by 300,000 steps at $L = 16$, 12 of 20 by one million at $L = 32$ and 8 of 20 at $L = 64$; given ten million steps, every world closed at $L = 16$ and 20 (Extended Data Fig. 5).

Closure is selected where it is found. Seeded at 1% into a world filled with the pusher, the evolved return closer at $L = 16$ and the block-copy tiling at $L = 32$ held half the cells within 400–750 and 260–290 steps and 0.64–0.91 of them at 20,000 steps, with every sampled heritable cell confined, in 5 of 5 soups each; the pusher seeded into a closer's world vanished in 5 of 5 (Extended Data Fig. 9d,e). Closure came sooner where the open phase ran unimpeded: the first closed replicator appears at a Kaplan–Meier median of 22,500 steps at $L = 16$ with benign tar (Stages G and L), against 38,000 with lethal tar, where there is no open phase (log-rank $P = 0.007$; Methods). The comparison changes lethality as well, and a dial between the two rules (below) shows that a damaged open phase delays closure further, so the open phase's part in closure is not settled; whether closers descend from the open form or displace it, these data do not say.

At $L = 50$ the most common tapes suggested closure and random cells withdrew it. The most heritable common tape, the pusher tiling with three inserted relative jumps, never enters its partner, but its offspring carries a fourth jump and does, and the lineage stops copying at the fourth transfer (Methods). The jump spreads, to 70–76% of the cells of evolved worlds and, seeded at 1% into its ancestor's world, to 73–75% within 300 steps; yet only 18–39% of heritable cells in the evolved worlds, and 19–25% of cells in the seeded ones, are confined (Fig. 4d; Extended Data Fig. 9a). At $L = 50$ the closed tape is a stage its lineage passes through, not an individual.

Where closure did not come, the closer was missing from what evolution had found, not from what the instruction set allows. In the 8080-like subset no world closed at $L = 32$ in a million steps (below), yet a 20-byte closer of 8080 instructions exists; seeded at 1% into a pusher world, its loop was in half the cells within 300–320 steps and the pusher fell below 1% within 490–1,000 steps in 5 of 5 soups (Extended Data Fig. 9b,c). The harness, too, supplies part of what closers use: the evolved closers we tested die when the registers are randomised (Extended Data Fig. 11a). With all registers drawn at random before each encounter, closure came in 1 of 60 worlds by 300,000 steps and 8 of 40 by three million, through closers that load their own source and offset, execute a jump, leave five to seven bytes unexecuted and copy with block-move instructions; in 6 of the 8 worlds they copy the whole tape, transmitters carrying a genome (Extended Data Fig. 10g; Methods). Four bytes suffice for a closer when the registers start empty and six when they do not, and without that help closure took more than a hundred times as long. A 13-byte self-initialising closer seeded at 1% into a random-register pusher world displaced the pusher within 650–1,300 steps in 5 of 5 soups (Methods). In both cases the barrier was discovery, not expressibility or competition.

Closure regenerates or transmits

Closure need not cost a lineage its variation, but in most worlds it did. We followed every single-byte mutant of 14 genomes through eight serial transfers into fresh random partners, counting every lineage as lost, alive with the mutation erased, or alive carrying it (Fig. 4a–c; Methods). The pusher's mutations reach its first copies, at 16 of 50 and 30 of 64 positions, but not their descendants, because 96–100% of its lineages stop copying within eight transfers, mutated or not; our registered prediction that its sites persist failed (Methods). The evolved closers, the return at $L = 16$ and the block-copy tilings at $L = 20$ and 32 , lose no unmutated lineage, but they copy the code they run and rebuild the rest of the tape from it: 60–88% of their mutant lineages live on with the mutation erased, and none keeps it. A third design, a short copying core followed by bytes it copies but never runs, keeps the allele in 46–63% of its mutant lineages after eight transfers, at 9 to 20 positions, 7 to 12 of them never executed (Fig. 4c). Such bytes are copied without being interpreted, half of what von Neumann's and Langton's descriptions do^{28,29}. Erasure is not canalisation in Waddington's sense³⁰, which buffers the phenotype while genetic variation persists: here the variation does not reach the offspring.

One byte decides which. A four-byte block-copy core recurs across stages, `XX 5e ed b0`: one instruction, then `LD E, (HL)` and the repeating block move `LDIR`; with its variant `1e NN ed b0` it is carried by 2,347 of the 15,655 confined heritable cells we scanned, and by most block-copy cells at $L = 32$ (Methods). Because the registers start at zero, the first byte is both an instruction and the copy offset: each byte is copied $d = XX \bmod 2L$ addresses ahead, and the block move, which outlasts the budget, tiles the first d bytes around the ring, the organism included (Fig. 6c). With $d < L$ the closer rewrites its own body from its first d bytes and erases every mutation beyond them; with $d = L$ it copies the tape verbatim and keeps them. The number of sites a closed lineage can transmit is therefore $d - 4$, set by one byte of its own code, whenever d divides L (Proposition 8). Of 2,343 confined block-copy cells sampled from every stage with such an offset, the single-mutant scan finds exactly $d - 4$ sites in 2,289 (98%), and the three cells whose offset does not divide L

carry fewer (Fig. 6c; Methods).

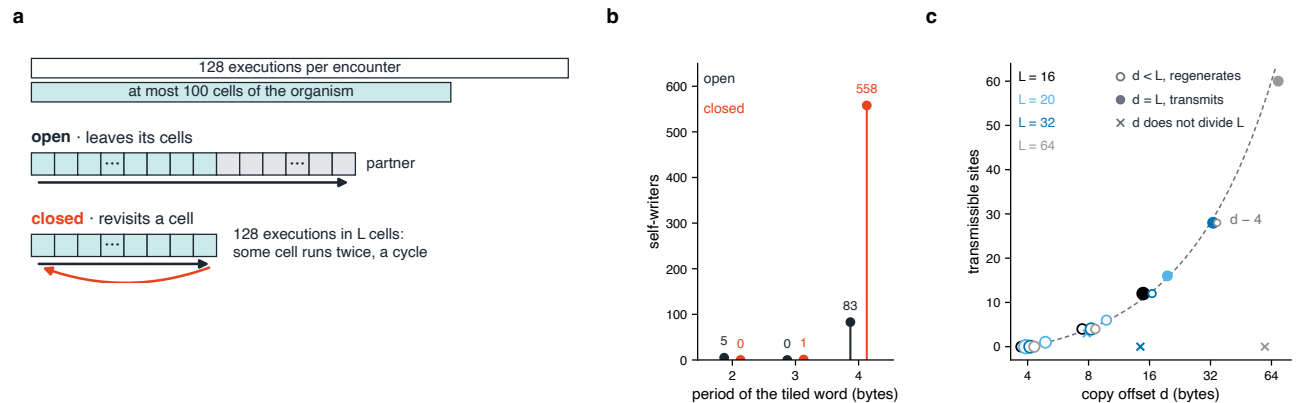


Fig. 6 | What is proved and what is counted. **a**, Theorem 2 in the form that holds for the Z80 soups with benign tar. An encounter executes 128 instructions, an organism has at most 100 cells (bars to scale), and no instruction stops execution inside it, so a program whose pointer never leaves its own cells (closed) must execute some cell twice, a cycle, whereas a program without one runs on into the partner (open). Under lethal tar a zero byte can stop execution inside the organism, and the argument does not apply. In BFF the same conclusion follows for a confined copier from the cost of a copy, at least 128 executions against a partner that differs from the tape in every cell (Theorem 1). **b**, Proposition 3: every word of period two, three and four, tiled to 16 bytes and executed against the all-zero partner; self-writers (the partner ends holding the tiling) by period, open (the pointer fetches a partner byte against the zero partner or one of 16 random partners) or closed. The five of period two are the load–push words; the one of period three, `b0 5e ed`, and all 558 closed ones of period four are block copies. Closed self-writers are the majority at periods three and four, so the open word’s head start, Kaplan–Meier medians of 450 steps to the first replicator and 21,000 to the first closed replicator in Stage G at $L = 16$, is not explained by counting. **c**, Proposition 8, the copy period: in the block-copy core `XX 5e ed b0` the first byte sets the copy offset $d = XX \bmod 2L$, the closer tiles its first d bytes over organism and partner, and the $d - 4$ bytes between the core and the offset are transmitted. Points, the median number of transmissible sites (single-mutant scan) of confined heritable cells carrying the core or its variant `1e NN ed b0`, by tape length and offset, for all 2,346 cells with $d \leq L$; the interquartile range is zero in every group with d dividing L , and 2,289 of the 2,343 such cells lie exactly on $d - 4$; crosses, the three cells whose offset does not divide L , which Proposition 8 does not cover; open circles, $d < L$ (regenerators); filled, $d = L$ (transmitters); dashed, $d - 4$. Marker area grows with the number of cells (1 to 1,121); small horizontal offsets separate the lengths. Proofs, the closure-window model and the lemma on information inflow are in the Supplementary Information. Source: `results/review_r2/census_p2.csv`, `census_p3.csv`, `census_p4.csv`, `results/offset/offset_cells.csv`, `results/offset/COPY_PERIOD_LAW.md`.

Which setting a world holds depends on the world, and its most common tape hides it. A regenerating population is nearly clonal, so its genome is the most common tape; a transmitting population is a cloud of distinct genomes, none of them common. We therefore classified random cells (Fig. 4d,e). Under benign tar, regenerators are the majority of heritable cells at the last snapshot in 17 of 20 worlds at $L = 16$ and 14 of 20 at $L = 20$, in 9 of 10 and 8 of 10 after ten million steps, and in 20 of 20 in the 8080 subset at $L = 16$; transmitters are the majority in 2 worlds at $L = 16$, 3 at $L = 20$ and 2 at $L = 64$. Under lethal tar, transmitters are the majority of heritable cells in 9 of 10 worlds, with a median of 11.5 sites they never execute, while the most common tape holds 0.2–1.5% of the soup: life that begins closed under lethal tar begins with a genome. So does most closure under random registers (above). In the ten-million-step worlds at $L = 20$, transmitters rose to a majority of heritable cells in 4 of 10 worlds before regenerators returned in 8 of 10 (Fig. 4e).

The mode belongs to the parent; the environment chooses among parents. Copied serially in six environments, under benign or lethal tar, with random partners or partners drawn from open, closed and lethal soups, and with random registers, each closer kept its mode wherever its lineages survived: the two regenerators carried no site and the three transmitters 9 to 12 in all 24 such combinations, while the environment decided survival, the return closer dying under lethal tar and every closer that relies on empty registers dying under random ones (Extended Data Fig. 11a; Methods). A matched regenerator and transmitter, with the same core and differing only in the offset byte and in what follows the core, cannot be told apart in encounters: each copies into every partner drawn from the soups, and survival as a partner, an intact core and viable single mutants differ between them by at most 0.01 (Extended Data Fig. 10e). Without mutation, their mixtures drift to either side (Extended Data Fig. 10d). With mutation, populations converge from either side to a share that the tar sets, at one length: at $L = 32$ transmitters hold 0.03–0.37 of the core-carrying cells after 300,000 steps with benign tar (medians 0.06–0.08) and 0.55–0.87 with lethal tar (medians 0.76–0.78), whether they start at 1%, 50% or 99% (Extended Data Fig. 10a,b). At $L = 16$ with lethal tar they hold 0.71–0.92 by 20,000 steps from every start, although the core was later lost in 7 of 20 worlds (Extended Data Fig. 10c). The earlier contest at $L = 32$ with benign tar, which the evolved regenerator won against a payload-carrying 8080 closer in 5 of 5 soups (Extended Data Fig. 9c), is one instance. Single mutants of the pair, followed in isolation, behave alike under both tars, so the choice is made in the population. Exploratory censuses find

one asymmetry: when a foreign executor's pointer runs into a regenerator, it executes the copies of the copying code that make up the regenerator's body and scrambles it, keeping its class in 0.05–0.24 of such encounters against 0.35–0.49 for a transmitter, whose body is junk; a transmitter whose body is tiled code is as vulnerable as a regenerator, and lethal tar widens the gap (Methods). This favours transmitters under either tar, so the full mechanism of the choice remains open.

Robustness, as usually measured, does not explain the succession. The first replicator is at least as robust to single mutations as its successor (0.74 against 0.59 of single mutants remain heritable at $L = 16$, 0.99 against 0.95 at $L = 50$; Extended Data Fig. 6), so a robustness advantage³¹ does not explain why closers replace it. The two kinds of closer are robust in different ways: a transmitter tolerates a mutation and passes it on, a regenerator survives it by erasing it, and only tolerance builds variation³². Cicala et al., who showed that a block-copy core can carry arbitrary trailing bytes, attributed the block-copy takeover to robustness, scored as an exact self-copy after random mutations²². By that criterion the transmitter of the matched pair scores 0.88 after one mutation and the regenerator 0.000, while our pushers and block-copy finals at $L = 32$ both score about zero (Extended Data Fig. 10f; Methods): the criterion measures the transmission of mutations, not survival.

Two properties of the substrate decide the beginning

If the order of events follows the literal channel, a machine without one should be born loop-bearing, and giving it one should start life open. BFF, the Brainfuck family of Agüera y Arcas et al.¹⁶, has no literal channel: its instructions increment in place or copy from memory, and copying a 64-byte tape into a partner that differs from it in every cell takes at least 128 instructions, 64 that change a cell and 64 that move a head, so a copier whose pointer stays in its own 64 cells must execute some cell twice (Theorem 1). In soups of 2^{17} programs run for 16,384 epochs (Fig. 5), BFF as published and with a wrapping pointer produced a heritable replicator in 28 of 48 worlds; every first replicator contains a loop (28 of 28), and none enters its partner in more than 34% of encounters (22 never do). With harmless brackets 7 of 12 worlds did, all loop-bearing but not all confined: the absence of a literal channel predicts a loop, not confinement. Adding one instruction that writes its own two operand bytes below the write head changes the beginning completely: in 12 of 12 worlds under either pointer rule the first replicator is the one-byte tiling of that instruction, loop-free, with its pointer entering the partner in every encounter (Fig. 5b). It is the 01 c5 of BFF, and without a wrapping pointer its offspring carry 8 bits of the partner, the ceiling of the measurement (Extended Data Fig. 7).

Then it dies. An unmatched bracket ends BFF execution, so the soup fills with tar that halts the pointer, copies into such partners are cut short, and by epoch 1,024 the open replicator holds under 1% of the soup; no closed class evolves, and none exists among periodic tilings of the literal, the brackets and a no-op up to period 10 (Methods). The open phase is a *window* whose length the tar sets. With harmless brackets the open replicator holds for all 16,384 epochs in 12 of 12 worlds; with brackets that halt with probability p it holds 98% of the soup at $0 < p \leq 0.1$ (18 of 18 soups) and collapses within about 900 epochs at $p = 0.3$ (6 of 6; Fig. 5c). When the literal writes two or three copies of its word, so that a copy completes within one pass, the open tiling's offspring carry 0 bits of the partner, information-closed from birth without a cycle, and the soup stays heritable: every random tape is heritable at 16,384 epochs in 12 of 12 soups (Fig. 5d). The lethal column holds in the first machine too. When a zero byte halts the Z80, its own tar becomes lethal and the open beginning disappears: in 10 of 10 pre-registered worlds no literal pusher establishes, and life begins closed and late, as a block copier, at a Kaplan–Meier median of 38,000 steps against 450 with benign tar (Extended Data Fig. 8), mostly as transmitters (above). Between the two rules lies a threshold, as in BFF. When a fetched zero halts the pair with probability p , the first replicator is an open load–push word in 10 of 10 worlds at every p up to 0.1, in 2 of 10 at $p = 0.3$ and in none at $p = 1$, and the share of transmitters among heritable cells at 300,000 steps rises with p , from a median of 0 at $p \leq 0.03$ to 0.26, 0.56 and 0.78 at $p = 0.1, 0.3$ and 1 (Spearman $\rho = 0.50$, permutation $P < 10^{-4}$; Extended Data Fig. 11b,d). Closure timing is not monotone in p : the first closed replicator comes at Kaplan–Meier medians of 18,000 steps at $p = 0$, 42,500–75,000 at $p = 0.01$ –0.1, where the open phase runs but is damaged, and 52,500–57,500 at $p = 0.3$ and 1, although the registered endpoint misses closed transmitter populations and so understates closure at high p (Extended Data Fig. 11c).

The beginning is not the Z80's alone. In an 8080-like subset, made by turning every Z80-only instruction into a no-op, which keeps the literal word but removes the block copies and relative jumps (Methods), the load–push word comes first in every world. At $L = 16$ the Z80's own return closer, pure 8080 code, dominates 20 of 20 worlds by 300,000 steps; at $L = 32$, where the return's popped addresses fall in the partner, none of 20 worlds closes in a million steps (Extended Data Fig. 5), as on long 8080 tapes in ref. 16, although a closer exists and, seeded, takes over (above). Together these fill a map (Fig. 5e): a literal channel decides whether life starts open; the lethality of the tar decides whether an open beginning lasts, or happens at all; and the closed designs within reach decide whether what follows is closure, a late closed beginning, or extinction.

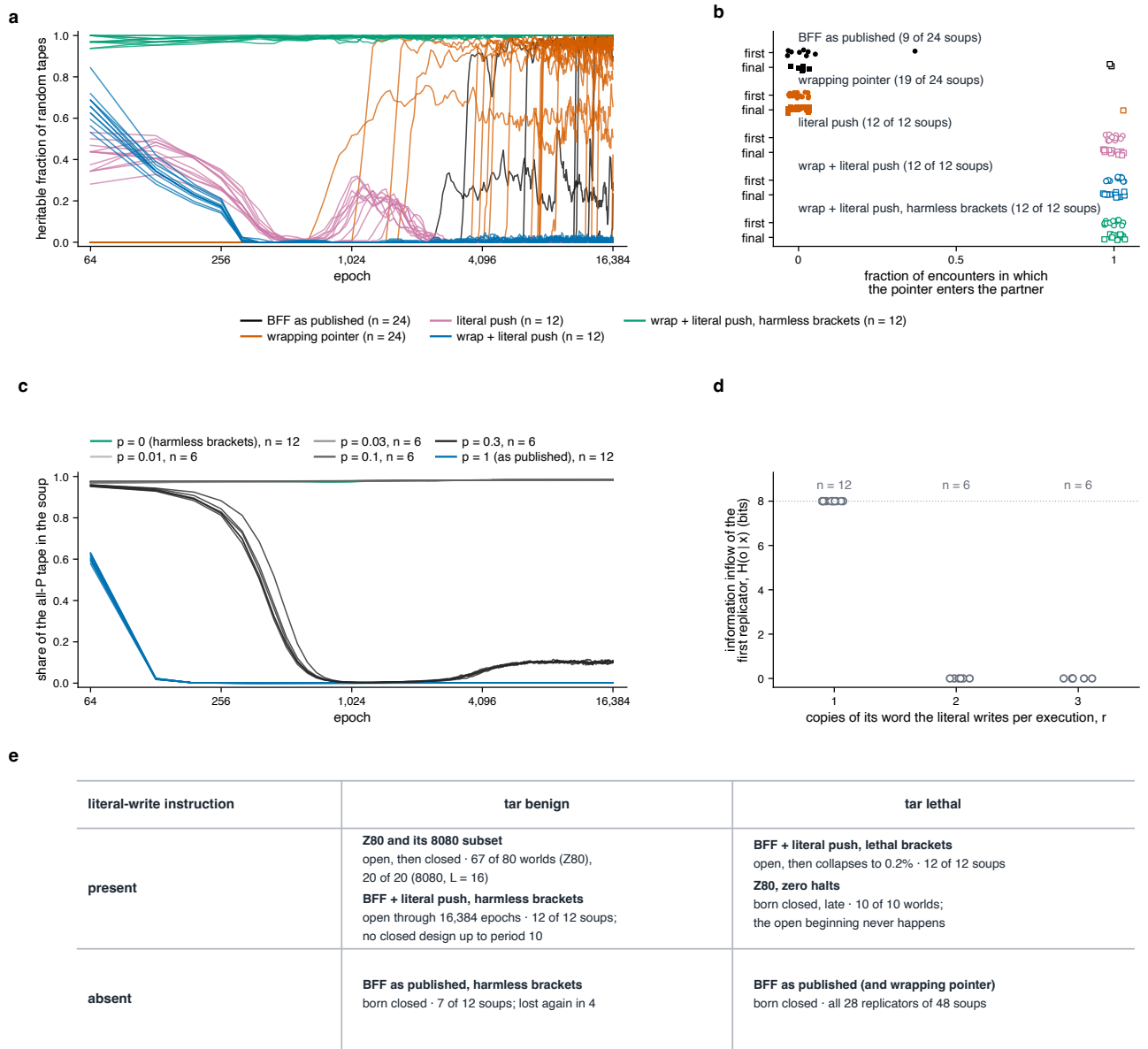


Fig. 5 | Two properties of the substrate decide the beginning. **a**, Heritable fraction of 32 random tapes against epoch for every BFF soup, by variant: as published ($n = 24$), with a wrapping pointer (24), with the literal push (12), with both (12), and with both and harmless brackets, which are no-ops instead of halting the pointer (12); samples every 64 epochs from epoch 64. **b**, Fraction of culture-test encounters in which the pointer enters the partner, for the first replicator (circles) and the final dominant (squares) of every soup with life, by variant (open symbols, no loop instruction; filled, loop instruction; jittered). **c**, Lethality as a dial: share of the all-P tape, the literal instruction tiled, against epoch under wrap + literal push when an unmatched bracket halts the pointer with probability p ($p = 0$, 12 soups; 0.01, 0.03, 0.1 and 0.3, six each; 1, as published, 12); the curves for $p \leq 0.1$ lie on top of one another near 1. **d**, The write ratio: information inflow of the first replicator, $H(o | x)$ over 256 random partners (dotted line, the 8-bit ceiling), when the literal push without a wrapping pointer writes $r = 1, 2$ or 3 copies of its word per execution; one point per soup, every first replicator loop-free. **e**, The map: the observed order of events in each cell defined by a literal-write instruction (present or absent) and by the lethality of the tar to an open replicator, with world counts; Z80 closure is counted at the population level (most heritable random cells confined at the last snapshot; Fig. 4d).

What is proved

Two counting statements bound the order of events without forcing it (Fig. 6; Supplementary Information). A straight-line program's pointer runs on into its partner, so a straight-line replicator is open by construction. A closed replicator, whose pointer never leaves its own cells, must execute some address twice wherever execution cannot stop inside it before the budget is spent, as in the Z80 soups with benign tar (128 executions, at most 100 cells), or wherever a copy costs more executions than the organism has cells, as in BFF, where copying a 64-byte tape into a partner that differs from it in every cell takes at least 128 (Theorems 1 and 2; Fig. 6a). Neither proviso holds under lethal tar, where a zero byte can stop execution inside the organism. Counting complete self-writers, the usual estimate of how likely a replicator is to arise by chance³³, does not explain which comes first (Fig. 6b). The five self-writers of period two are all open load-push words, the single one of period three, b0 5e ed, is a closed block copier, and of the 641 of period four, 558 are closed block copiers (Proposition 3). Closed self-writers are thus the majority one period above the smallest, and the open word's head start, Kaplan–Meier medians of 450 steps to the first replicator and 21,000 to the first closed one at $L = 16$, is measured, not counted. Theorem 2 concerns confinement, and the second machine shows that independence from the partner can be had without it: the BFF literal that writes two or three copies of its word enters the partner in every encounter, yet its offspring carry nothing of the partner (Fig. 5d), information closure without pointer closure (Lemma 7). The two coincide in the Z80's loop-bearing finals, whose measured inflow falls to zero (Fig. 3c); independence, zero inflow from context to offspring, is what perfect fidelity requires (Lemma 7). The dynamics lie outside the theorems: closure arrives with probability about $1 - \exp(-q \int n dt)$, q being the rate of closing mutations and n the open population, so benign tar lets the window run until a closer is found and lethal tar shuts it first, a reading that the lethality dial supports only at its two ends (Supplementary Information, Model 5).

Discussion

The order of events is observed, and the counts bound it: the smallest self-writers of this machine are open, a closed one needs a return, and the open word's head start is measured rather than derived. So the first replicator is open; closure is a return of control flow into the organism, by a conditional return, a repeating block move or a jump, found where a closing design is within reach and selected once found; and closure is not one thing for heredity. A closer that copies only the code it runs regenerates itself and erases its mutants; a closer that also copies bytes it never runs keeps their variants through serial transfers; in a recurring block-copy core one byte, the copy offset, decides which and sets how many sites a lineage can transmit. Benign-tar worlds that closed at $L = 16$ and 20 ended mostly with regenerators, and lethal-tar worlds, like most random-register worlds that closed, with genomes; a matched pair at one length shows that the tar, not the design, makes that choice, and the lethality dial that the share of genomes rises with it. The explanation has few adjustable parts: the literal channel is read off a substrate, the copy offset off a genome, and the lethality of by-products is measured. It answers questions it was not built for: why the first replicators are sloppy; why a cycle, not a membrane, is the first thing selection builds; why inherited variation can fall at the first transition in individuality, when the first closer regenerates; and why the first innovations after replication should concern where copying starts, stops and returns. When the literal road is blocked, by removing the stack writers, by a language without a literal channel or by lethal tar, life can still come, late and loop-bearing.

These are predictions for chemistry, not results about it. Template copying is a literal channel, a strand specifying its complement by physical fit with no interpreter, so the first replicators of such a chemistry should be the shortest self-templating repeats, dependent on their surroundings and productive mainly of partial copies, which transfer into fresh material would show. By-products are a variable: whether a chemistry's tar is inert or inhibitory to copying should decide whether an open phase lasts long enough for control to evolve, which matched inert and inhibitory additives in nonenzymatic systems can test²⁴. Processivity is the chemical write ratio, which in the second machine closes a copier in information without any control: vertical lineages should appear when one copying event spans the genome³⁴, and in laboratory evolution of RNA replicators the earliest fixed changes should concern control, circularisation, terminators and primers^{35,36}, and inherited sequence that is copied but not used should be commoner where by-products are inhibitory. The copy offset has a chemical counterpart, the point at which a copy restarts on its own template: a restart within the copied unit regenerates a short repeat, a restart at the unit's end transmits the whole of it. Detection of life at its origin should be by intervention, since low-complexity order precedes the first replicators and trips compression and assembly measures alike^{25,26}. What limits the claims is plain: the Z80, an 8080-like restriction of it, and one toy language; units of fixed size, start address and pairing imposed by the harness; a culture test that is our own ground truth, though the open beginning held under both register conventions we tested, at every length from 16 to 64 bytes and under heredity thresholds from 0.2 to 0.4; closure at the largest lengths in progress rather than complete within our horizons, absent from populations at $L = 50$,

where the closed tape is a passing stage, and a hundred times slower when the harness does not empty the registers; and an environmental choice between the two kinds of closer whose mechanism we have not found. Within those limits the result stands: heredity came first; independence from context came later, as a return in control flow; and whether that return cost the organism its variation depended on whether it regenerated the code it ran or also copied bytes it never ran, a choice one byte of the genome encodes and the environment selects.

Methods

Soup. The soup holds $N = 20,000$ tapes of L bytes (L from 3 to 100), initialised with uniformly random bytes, on a 160×125 square lattice. At every step 8,192 cells are drawn uniformly at random and each is paired with one of its four lattice neighbours (a step off the edge is reflected; the lattice does not wrap); a pair is active if neither member is already in another pair that step, which leaves $\approx 3,980$ active pairs. An active pair is concatenated, organism first, into a pair memory of P bytes ($P = 2L$ except in the padding experiment of Stage F, where $P \geq 2L$ and the padding bytes are never written back), executed by a Z80 emulator for a fixed budget of 128 instructions (32–2,048 in the budget arms) from byte 0 with all registers zero and the stack pointer at the last byte of the partner, and both tapes are written back. After each step $8,192/2^k$ bytes chosen uniformly at random over the soup are replaced by random bytes ($k = 4$ unless stated; $k \in \{2, 4, 6\}$ in the mutation arms of Stages A, C and D and $k \in \{1, 2, 3, 5, 8\}$ in the Stage E mutation sweep at $L = 100$; the per-byte-constant arm of Stage E replaces $32 \cdot L$ bytes per step, the $L = 16$ rate per byte, at every length). There is no fitness function, no replacement and no death. Instruction-set ablations replace whole opcode families by no-ops through a prefix-aware fetch hook that sees the operation after `DD/FD` and `ED/CB` prefix resolution; the ablations and their opcode counts are listed in Extended Data Table 2. The emulator was validated against an independent Z80 reference, with and without ablation, by a differential harness that runs random programs through both (`src/lib/dev/z80difftest.ts`); no claim depends on cycle-exact timing.

Recording. Stages C–M write a sample at steps 1, 2, 3, 5, 8, 13, 21 and 34, then every 50 steps to step 5,000, then every 500 steps (300,000-step runs), 1,000 (one-million-step runs) or 5,000 (Stage L), with the ten most common exact tapes and their shares, quasispecies³⁷ occupancy (fraction of cells within Hamming distance $\lceil L/4 \rceil$ of the dominant tape), byte histogram, zero-byte fraction and byte entropy, brotli-compressed bits per byte and the high-order entropy of ref. 16, species entropy and unique-tape fraction, per-interaction statistics (bytes written into the partner, copy events, destroyed programs), 16 random tapes for the functional census, and full soup snapshots at fixed steps and at emergence. Stages A and B sample every 500 steps from step 500, with the three most common tapes and without the byte histogram, the per-interaction statistics or the random tapes.

Culture test and partner tests. A tape runs as the first program against 64 uniformly random partners for the condition's budget; copy success is the fraction of partners that become $\geq 75\%$ copies at the best cyclic shift. The score of a tape x is the mean, over partners j whose best-shift match to x before the encounter, s_j , is below 0.75, of $(s'_j - s_j)/(1 - s_j)$, s'_j being the match after it. `gen2` is the same quantity one generation on: the offspring of encounter j runs as organism against a fresh random partner, whose match to x is measured before and after, so offspring that are not copies contribute about zero and `gen2` approximates the share of encounters whose offspring copies in turn. Heritable means `gen2` ≥ 0.3 , faithful additionally $\geq 50\%$ of partners copied. `gen2` is a normalised gain in best-shift similarity to the original tape, not a count of copies, so a word that smears its return addresses into its partners raises their similarity to itself without copying: 246 two-byte census words pass `gen2` ≥ 0.3 without producing a single 75% copy (Extended Data Fig. 2). Replicators are therefore also scored by the copies they make (faithfulness and the partner tests), and dating the first replicator by the faithful criterion (`t_faith`, below) instead of by heredity alone gives the same tape at the same step in all 80 Stage G worlds. Partner-independence tests use 256 random partners and record self-damage ($\geq 25\%$ of the organism's bytes changed in one encounter). In BFF the culture test also records whether the instruction pointer entered the partner in each encounter; closed means $\leq 5\%$ of encounters entered and $\geq 95\%$ of partners copied, open means $\geq 50\%$ entered. Information inflow from the environment, $I(o; e | x) = H(o | x)$ for a deterministic executor, is the plug-in entropy in bits of the offspring (the partner's bytes after the encounter) over 256 random partners, ceiling 8 bits.

Events and statistics. `t_rep` is the first sampled step at which one of the three most common tapes has a share $\geq 0.5\%$ and is heritable by the culture test (a clonality criterion by design: it dates the first *clone*, not the first heritable material); `t_faith` additionally requires faithfulness; the *final dominant* is the most heritable of the three most common tapes at the horizon. It is the most common tape in 57 of the 60 Stage G worlds at $L = 16, 20$ and 64; at $L = 50$ it is the second most common in all 20, behind the plain pusher tiling (5.5–6.6% of cells), while the tapes that carry its jump word `20 f0` hold 70–76% of cells. Its offspring is then the parent with one more `20 f0` inserted, the same string in every encounter; that offspring enters the partner in every encounter, the next generation (six jumps) is confined but fails the culture test (`gen2`

0.19), and the one after (ten jumps) no longer copies (`150_lineage.py`), so at $L = 50$ the final dominant is a stage of a lineage rather than a lineage. Because the most common tape can misrepresent a diverse population, population claims rest on random cells (Population sampling, below); $\tau_{q_{10}}$ (quasispecies occupancy $\geq 10\%$) is the pre-registered occupancy event and fires on zero floods as well as on replicators. Emergence fractions are quoted as worlds reaching the event within the horizon with Wilson 95% intervals; times as Kaplan–Meier medians with censoring at the horizon. Ablation effects are ratios of Kaplan–Meier medians with seed-paired sign tests (the same seeds run under every ablation); confirmatory contrasts use one-sided Fisher exact tests and the Cochran–Mantel–Haenszel test pooled over budgets; between-variant transition rates in BFF use two-sided Fisher tests and Mann–Whitney tests on event times with non-transitioning runs censored at the horizon. Across the 233 unablated worlds of the exploratory stages, 92 gained a control-flow instruction between the first replicator and the final dominant and 3 lost one. In BFF, where replicator populations are quasispecies with small exact-class shares, the first replicator is the most common tape class (a tape and its reverse count as one class) at the first sample at which it is heritable, and the population event is a heritable fraction ≥ 0.5 of 32 random tapes. All thresholds (0.75 copy, $\text{gen2} \geq 0.3$, 0.5% share, 10% occupancy, 95%/5% closure) were fixed before the confirmatory stages. Re-dating the first replicator from the recorded runs under heredity thresholds from 0.2 to 0.4 leaves the load–push word first in 170 of 170 benign-tar worlds of 16–64 bytes; at 0.5 and 0.6, near the pusher’s own gen2 , it is first in 155 of 169 and 107 of 164. Principal medians are given with 95% bootstrap intervals across worlds (Extended Data Table 3). Same-seed Z80 runs diverge because GPU scheduling makes the pairing order non-deterministic, so every Z80 claim is a population claim over seeds (ten same-seed repeats spread as widely as ten seeds); the BFF soup is bitwise reproducible.

Definitions and theorems. A *self-writer* is a tape that, executed against the all-zero partner, leaves the partner holding a copy of itself. *Closure* (confinement): the organism’s pointer never fetches a partner address as instruction in any encounter with a partner of the sampled class (uniformly random partners). Theorem 1: copying a 64-byte BFF tape into a partner that differs from it in every cell takes at least 128 instructions, 64 that change a cell and 64 head moves, so a program whose pointer stays in its own 64 cells executes some address twice in that encounter, that is, runs a loop. Theorem 2: in any machine whose pointer advances by default, a closed full replicator executes some address twice, provided a copy costs more instruction executions than the organism has addresses (BFF) or execution cannot stop inside the organism before the budget is spent (the Z80 soups with benign tar, budget $128 > L$; not Stage I, where a zero byte halts, nor the Stage E arms with budgets 32 and 64 at $L = 100$). Theorem 6: a self-consistent literal-write instruction gives a straight-line, hence open, replicator the size of the literal; without one, under the copy-cost hypothesis (H2), which we do not prove for BFF, and in a machine whose execution ends when the pointer leaves the pair, every self-writer executes some address twice. Lemma 7: $I(o; e | x) > 0$ implies imperfect fidelity across contexts; perfect fidelity implies $I = 0$. Proposition 8: in the block-copy core `XX 5e ed b0` with copy offset $d = XX \bmod 2L$ dividing L , every encounter leaves organism and partner holding the d -periodic tiling of the organism’s first d bytes, so exactly the $d - 4$ bytes between the core and the offset are transmitted. Proposition 3 (computer-assisted, `census_period.py`): every word of period two, three and four, tiled to 16 bytes and executed against the zero partner. Period two: exactly six self-writers, the trivial `00 00` and the five load–push words, all straight-line and open. Period three: one, `b0 5e ed` tiled, closed, a block copier whose copy offset equals the tape length. Period four: 641 of primitive period four, 558 of them closed, all block copies (455 `LDIR`, 103 `LDDR`), and 83 open. A self-writer is closed if its pointer fetches no partner byte against the zero partner and 16 random partners. Proposition 4 states what the count of the initial soup does and does not show. Full statements and proofs are in the Supplementary Information.

Harness conventions. Two derived soup and executor shaders, byte-identical to the Stage G ones except after the register reset of every encounter (`gen_conv_shader.py`): *random registers* draws A, F, B, C, D, E, H, L, their alternates, IX and IY uniformly (program counter 0, stack pointer as before); *random stack pointer* draws SP uniformly. Draws are seeded by the step’s batch seed and the pair index. Twenty worlds each at $L = 16$ under Stage G conditions for 300,000 steps (seeds 8001–8020), and 40 random-register worlds for 3,000,000 steps (seeds 8101–8120, and 8201–8220 as a registered replication after the first set closed only in odd seeds); every assay, including the culture test and tracing, runs under the world’s own convention. Closed: most heritable cells among 64 random cells confined in 16 traced encounters. Registered predictions and outcomes (REVISION_PREREG C and R4): the first replicator is a load–push word under random registers in 20 of 20 (met) and open under a random stack pointer in 18 of 20 (met); closure in at most 5 of 20 random-register worlds by 3,000,000 steps (not met: 6 of 20), and in 2–12 of 20 in the replication (met: 2, both even seeds). Closers of the closed random-register worlds were characterised post hoc (32 heritable cells each: whole-tape exact copy over 16 encounters, never-executed bytes by tracing). Invasions under random registers at $L = 16$: the 13-byte closer `21 00 00 11 10 00 01 10 00 ed b0 18 fe` with a random 3-byte tail per cell, seeded at 1% into a pusher-filled world, and the reverse, five seeds each and three unseeded controls, 50,000 steps; under the standard convention, the same closer against the 4-byte regenerator `44 5e ed b0` $\times 4$, 100,000 steps. The registered motif-share endpoints failed on definition (neutral operand bytes drift in a transmitter); classes of 64 random cells at the end are reported.

Invasions at aligned lengths. Stage G dynamics at $L = 16$: the return closer `ad e3 21 e3 21 c0 ad c0` $\times 2$ against the pusher `01 c5` $\times 8$; Stage K dynamics at $L = 32$: the block-copy tiling `04 5e ed b0` $\times 8$ against `01 c5` $\times 16$. Each seeded at 1% into a world filled with the other, five seeds each way, three unseeded controls per resident, 20,000 steps; class share (within Hamming distance $L/4$ of the design at any shift) every 10 steps to 500 and every 250 after, and classes of 64 random cells at 1,000, 5,000 and 20,000 steps. Registered: the closer holds most cells and most heritable cells are confined at 20,000 steps in at least 4 of 5 soups at each length (met, 5 of 5 at both), and the pusher stays below 5% in a closer world (met, 5 of 5).

First closure. For every world of Stages G, H, I, K, L and M, the first sample at which one of the three most common tapes holding at least 0.5% of cells is heritable and confined in 16 of 16 traced encounters, censored at the horizon; Kaplan–Meier medians and a log-rank test of benign $L = 16$ (Stages G and L, 30 worlds) against lethal $L = 16$ (Stage I, 10 worlds), with the decision rule registered before the analysis (`first_closure.py`).

The copy-offset switch. In the block-copy core `XX 5e ed b0` with $HL = D = B = C = 0$, register E is loaded from the organism's first byte, so `LDIR` copies each byte $d = XX \bmod 2L$ addresses ahead; for `1e NN ed b0`, $d = NN \bmod 2L$. The working values of `XX` were found by executing the core with every byte value (`offset_census.py`). The copy-period law pools every confined heritable cell of the population scans whose first four bytes are either core and compares d with the number of transmissible sites of its single-mutant scan (`offset_population.py`). The matched pair (REVISION_PREREG E): at $L = 16$, $R = 44 5e ed b0 \times 4$ and $T = b0 5e ed b0$ followed by 12 random bytes drawn for every cell; at $L = 32$, $R = 04 5e ed b0 \times 8$ and $T = a0 5e ed b0 + 28$ random bytes. Conditions $L = 16$ with benign tar, $L = 16$ with lethal tar and $L = 32$ with benign tar; starts 50:50 (ten worlds without mutation for 100,000 steps and ten with the standard mutation for 300,000 steps) and 1% each way with mutation (five each); every cell classified by its first four bytes every 50 steps to 5,000 and every 500 after (`offset_switch.py`, Modal). The registered predictions of a mutation-bias equilibrium were not met (E1, E2); without mutation the share drifts. Encounter components of the pristine pair were measured with 4,096 encounters each against cells of the final soups (`offset_encounter.py`). Mutational flows (`offset_flow.py`, registered as E-mech): every single-byte mutant of R and T , four random partners each, classified by whether its offspring copies itself exactly and with what period; the registered prediction (EM1) failed and the flows of isolated mutants are the same under benign and lethal tar, so the environment dependence is reported without a mechanism.

Encounter censuses (exploratory). Lattice-neighbour pairs (200,000 per soup set) from the final soups of the ten 50:50 $L = 32$ worlds of each tar, executed under each rule (composition crossed with rule), with every cell classed by its first four bytes before and after (`offset_census_enc.py`); and core-free executors from the same soups against three partners, R , T with a random body and T with a tiled body (`a0 5e ed b0 \times 8`), traced to see whether the executor's pointer entered the partner (`offset_intruder.py`, 16,384 encounters per combination). Both were run after the matched-pair results and are reported as descriptive.

Parent $\times$ environment. Six parents at $L = 16$ (the pusher; the evolved return closer; R and T of the copy-offset switch, T with a fixed random tail; the evolved lethal-tar closer of world 4009; the self-initialising closer `2e 00 1e 10 ed b0` with a fixed random tail) were followed through eight serial transfers, with every single-byte mutant in 16 lineages and 256 unmutated lineages, as in the serial-retention test, in six environments: benign or lethal rule with uniform random partners; partners drawn from the final soup of Stage G world 2001 (closed) or from its snapshot at 5,000 steps (open phase), benign rule; partners from the final soup of Stage I world 4001, lethal rule; random registers with random partners (`pe_matrix.py`; registered as PE). Registered: wherever a parent's lineages survive, regenerators carry at most 2 sites and transmitters at least 8 after eight transfers (met, 24 of 24); the pusher's lineages are lost everywhere (not met: 0.36 alive with partners from the closed world, with 0 sites).

Lethality dial. Derived soup and executor shaders (`gen_dial_shader.py`) in which a zero byte fetched as the first byte of an instruction halts the pair with probability p , a fresh draw from a per-encounter generator for every such fetch; $p \in \{0, 0.01, 0.03, 0.1, 0.3, 1\}$. Before any soup ran, the $p = 0$ executor reproduced the standard executor and the $p = 1$ executor the Stage I executor exactly on 65,536 random pairs with 25% zero bytes. Ten worlds per p at $L = 16$ under Stage G conditions, the same seeds (9001–9010) at every p , 300,000 steps; top-3 tapes every 50 steps to 2,000, every 250 to 20,000 and every 2,500 after; every assay under the world's own p (`dial_soups.py`; registered as DZ). Registered: the first replicator is open in at least 8 of 10 worlds at $p = 0$ and at most 2 at $p = 1$, with a falling trend (met: 10 of 10, and 0 of the 9 worlds with a heritable common tape; Cochran–Armitage $z = -6.1$); first closure later at higher p , monotone up to one reversal and Spearman-positive (Spearman met, $\rho = 0.25$, permutation $P = 0.023$; monotone part not met); the transmitter share of heritable cells rising with p (met, $\rho = 0.50$, $P < 5 \times 10^{-5}$) with transmitter-majority worlds at most 3 of 10 at $p = 0$ and at least 7 at $p = 1$ (1 and 6: the second not met). The registered closure endpoint (a top-3 tape holding 0.5% of cells) misses closed transmitter populations, whose genomes all differ: two worlds counted as unclosed ($p = 0.03$ and 1) are closed and transmitter-dominated at 300,000 steps, which biases the timing test against closure at high p . The closing-core counts

(2,347 of 15,655 confined heritable cells carry XX 5e ed b0 or 1e NN ed b0) are from `core_families.py`.

Ref. 22's robustness criterion. Every Stage K first replicator and final dominant ($L = 32$): 1,000 trials each of 1, 4 and 8 successive random single-byte mutations; a trial counts if the mutant, run for 512 instructions against the all-zero partner, leaves the partner equal to the mutant. Registered: block-copy finals more robust than pushers at every k (not met: medians 0.000–0.002 for both). The matched pair of the copy-offset switch was scored the same way post hoc (`robust_ref22.py`).

Traced execution. A derived copy of the culture-test executor records every address fetched as instruction stream (opcodes, prefixes, operands) in a 256-bit map per encounter, with byte-identical execution (checked). Pointer confinement is the fraction of 256 encounters with a seeded random partner in which no partner address is fetched; the executed set is the union of fetched organism positions; a transmissible position of the mutational scan is called unexecuted if it is never fetched. Every first replicator and final dominant of Stages G, K and I (110 of each) was traced. Confinement concerns the instruction pointer only; a confined copier may read its partner's bytes as data.

Mutational scan. For each Stage G first replicator and final dominant, every single-byte mutant (all 255 alternative values per position at $L = 16$ and 20; 32 seeded values per position at $L = 50$ and 64) was run as organism A against 32 shared random partners for 128 instructions and its offspring against 32 fresh partners, with the culture test's thresholds. A mutant is heritable if $\text{gen2} \geq 0.3$; its mutation is transmitted if the mutant byte is present at the aligned position in at least half of the offspring that are $\geq 75\%$ copies; a position is a transmissible site if at least half of its tested values are both heritable and transmitted; the single-mutant variation score is $\sum_i \log_2(1 + 255 f_i)$ over positions, f_i being the fraction of transmissible values at position i , a summary of how many single mutants are heritable and reach first-generation copies, not a channel capacity (it ignores epistasis and later generations). Transmission is scored in first-generation copies only; serial retention follows alleles further. Three predictions were written before the run (the successor more robust to mutation than the first replicator; the closed form with more transmissible sites; the matched pusher-with-jump of $L = 50$ with more sites than its pusher); all three failed and are reported as such.

Pair invasion. Soups of 20,000 cells at $L = 50$ under the Stage G dynamics, every cell set to one tape of the matched pair (the pusher tiling, or the same tiling with 20 f0 inserted three times, the final dominant of 14 Stage G worlds) and 1% of cells to the other at step 0; five seeds per direction, three seeds per resident with no invader; 20,000 steps. The pre-registered outcome, the invader's nearest-class share passing 50%, could not be met at this length because the quasispecies cloud splits between the two classes (a closed tape that has lost two jumps is nearer the pusher); the share of cells carrying the jump word, reported instead, was defined after the data were seen. In single-tape invasion assays the pusher tiling 01 c5 $\times$ 8 seeded into 1% of the cells of a random soup at $L = 16$ (128 instructions, mutation 1/16, five soups) peaked at a median occupancy of 0.32 at step 250. The jump-word marker was checked against traced confinement before use (registered): in three re-runs of the closed-into-ancestor invasion, 512 random cells at each of eight steps were traced against 16 random partners. The marker share reached 0.73–0.75 while the share of cells confined in all 16 encounters stayed at 0.19–0.25, and only 26–33% of marker-carrying cells were confined; the registered agreement failed, so the $L = 50$ invasion is reported as the spread of the jump, not of closure (Extended Data Fig. 9a).

Serial allele retention. Fourteen genomes (Fig. 4a) were scanned with every single-byte mutant (all 255 values per position at $L \leq 32$, 32 seeded values at $L \geq 50$): four open pushers, the most common closed final dominant at each of $L = 16$, 20 and 32, the $L = 50$ final dominant, the lethal-tar final dominant with the most first-generation sites, the most common tape of three Stage L worlds at the first snapshot at which a genome-bearing block copier was most common, and the 20-byte 8080 closer with two random 12-byte payloads. Each mutant founded 16 lineages and the unmutated tape 256. In each of eight transfers the lineage tape ran as organism against a fresh uniformly random partner under the genome's own rule (8080 subset, lethal tar) and the partner's bytes after the encounter became the next lineage tape; lineage r met the same partners for every mutant. A transfer is a copy if the offspring matches its parent at $\geq 75\%$ of positions at the best cyclic shift, and a lineage is alive while every transfer has been a copy. The allele is carried if the mutant's five-byte window centred on the mutation occurs anywhere in the lineage tape, cyclically, which needs no alignment, so inserted jumps and shifts do not hide it; mutants whose window occurs in the unmutated tape would be excluded (none did). Every lineage is counted as lost (not alive), erased (alive, allele absent) or carrying the allele. A position is serially transmissible after g transfers if, for at least half of its values, the share of alive lineages carrying the allele exceeds by at least 0.5 the share of alive unmutated lineages that contain the same window by chance. Design and predictions were registered before the run: the pusher's sites were predicted to persist (failed: its lineages are lost); the evolved closers to erase, the transient genomes to keep at least five sites and the constructed closer its payload (met). A post hoc variant drew partners from cells of the organism's own world (Stage G, seed 2001) instead of random bytes.

Population sampling. At the final snapshot of every Stage G, I, K, L and M world, 256 random cells were culture-tested (32 partners) and traced against 16 random partners; a cell is confined if no partner address is fetched as instruction in

any of the 16 (registered). At every snapshot of the same worlds, 64 random cells were culture-tested and traced, and up to 16 random heritable cells were given the single-mutant scan (16 values per position, 16 partners) and their executed set (defined before the run and exploratory, since the first sample motivated it). A heritable cell is open if not confined; a confined one is a regenerator (at most two transmissible sites), intermediate (three or four) or a transmitter (five or more); the classes of unscanned confined cells are inferred from the scanned ones. A world is closed at the population level if most of its heritable cells are confined. For confined cells the first-generation scan agrees with serial retention (Fig. 4a,c); for open cells it overstates it, so open cells are reported as open.

Invasions of a constructed closer. Stage M dynamics at $L = 32$. A1: the 20-byte 8080 closer with payload 1 seeded at 1% into a world filled with the 8080 pusher, and the reverse, under the 8080 subset, five seeds each and three unseeded controls per resident, 20,000 steps. A2: the same closer against the evolved block-copy tiling `04 5e ed b0 × 8` under the full Z80, 50,000 steps. The registered outcome, the share of cells carrying the closer's exact 20-byte core, failed on definition: the core mutates within a few hundred steps while its descendants hold the soup. We report the share carrying the closer's loop body (post hoc) and the classes of 64 random cells per soup at the last step (Extended Data Fig. 9).

Address mapping and aligned lengths. Pair memory is a ring of $2L$ bytes: every memory access uses the 16-bit address modulo $2L$, and the stack pointer starts at the largest 16-bit value that aliases the last byte of the partner. These conventions, addresses taken modulo the pair length, the stack pointer at the end of the pair and every register empty at the start of an encounter, follow ref. 16, whose tapes were 16 bytes long, an aligned length; the evolved closers rely on the empty registers (Harness conventions). The program counter and the stack pointer themselves are 16-bit, so a relative jump or a stack move that crosses address 0 reaches $(65,536 + \text{target}) \bmod 2L$, which equals the true-ring target only when $2L$ divides 65,536. Of the 67 closed final dominants of Stage G, 25 carry a relative jump (JR or DJNZ) at a length where the 16-bit program counter does not fold consistently onto the ring: 20 at $L = 50$ and 5 at $L = 20$, one of which (seed 2003) is classed by its block repeat in Extended Data Table 1 (we did not check that every one of these jumps crosses address 0); the pair-invasion and mutational-scan results at $L = 50$ are results about the machine so defined. Stage K ($L = 32$, 20 worlds, one million steps, seeds 5001–5020; pre-registered: first replicator a load–push word in ≥ 18 of 20, a closed successor in ≥ 15 of 20, kill below 10) tested closure at an aligned intermediate length: 20 of 20 and 12 of 20, all closers block copies.

The 8080 subset. The `i8080` suppression makes every Z80-only instruction a no-op: the 256 CB-page opcodes, the 78 defined ED-page opcodes (block copies and their repeats, 16-bit ADC and SBC, IN and OUT (C), NEG, RETN and RETI, IM, LD A,I and A,R, RLD and RRD) and EX AF,AF', DJNZ, the five JR forms and EXX. This is an 8080-like restriction of the same emulator, not an 8080. Three deviations from a real 8080 are kept: the IX and IY prefixes `DD` and `FD` still select the indexed form of an 8080 instruction, where the 8080 executes them as CALL; the 8080's other alternate opcodes are no-ops here (CB, JMP on the 8080, and ED, CALL, become two-byte no-ops with the byte that follows, and `D9`, RET, a one-byte no-op); and the core keeps Z80 flag behaviour, so P/V holds overflow after arithmetic where the 8080 sets parity, and conditional jumps, calls and returns on PE or PO can differ. Stage M: Stage G conditions at $L = 16$ (300,000 steps) and $L = 32$ (one million), 20 worlds each, seeds 7001–7020, with the predictions registered before the run. The registered reading that the subset expresses no closer on a 32-byte ring was wrong and is withdrawn: a reviewer's 20-byte closer of 8080 instructions (`21 20 00 31 40 00 16 10 2b 46 2b 4e c5 15 c2 08 00 c3 11 00`, a counted push loop closed by a conditional jump, then a jump to itself) copies itself and any 12 following bytes exactly into 256 of 256 random partners without entering them, in our executor under the subset (`check_8080_closer.py`).

Ten-million-step runs. Stage L: the Stage G conditions at $L = 16$ and 20, ten worlds each (seeds 6001–6010), run for ten million steps with samples every 5,000 steps and full snapshots at the Stage G steps and at 2, 3, 5, 7.5 and 10 million. At every snapshot of these and of the Stage G, K and I worlds, the most common exact tape was scanned (32 values per position, 32 partners) and traced (64 partners), giving its transmissible sites, single-mutant variation score, executed set and pointer confinement against step (a modal-tape analysis; population classes are in Fig. 4d,e). Predictions registered before the run: L1, in at least 5 of 10 worlds at each length the dominant at ten million steps has more transmissible sites than the closed dominant at 300,000 ($L = 16$) or one million ($L = 20$) steps; L2, every such recovery is in a block-copy lineage and its sites are copied-but-unexecuted positions; L3, no return-based or push-based dominant with more than one transmissible site at any snapshot. Outcomes: L1 not met (1 of 10 at $L = 16$, 0 of 10 at $L = 20$); L2 not met at $L = 16$ (the one recovery, from 0 to 4 sites, is a block-copy lineage with one of its four sites unexecuted) and not applicable at $L = 20$; L3 not met as written, at $L = 16$ (the open pusher of seed 6003 holds two transmissible sites at eight snapshots from 2,000 to 30,000 steps) and at $L = 20$ (the open pusher holds two to three sites and the jump-based closer two); the kill criterion fired (final dominant with at most one site in 9 of 10 and 8 of 10 worlds). Every world closed, at 10,000–75,000 steps ($L = 16$) and 20,000–500,000 ($L = 20$); the heritable fraction of random cells at ten million steps is a median 0.89 and 0.95.

Controls. Stage H, well-mixed pairing (the partner drawn uniformly from the soup) at $L = 16$, ten worlds, seeds 3001–3010, against the 20 Stage G worlds at $L = 16$. Pre-registered: H1, the open phase less heritable under mixing (killed: the

median heritable fraction of 16 random cells over steps 2,000–20,000 is 0.125 in both, one-sided $P = 0.78$); H2, closure sooner under mixing, closure here meaning `t_close70`, the first sampled step at which the heritable fraction of 16 random cells reaches 0.7, not pointer confinement (killed: median 50,000 against 30,000 steps, reached by 300,000 steps in 7 of 10 worlds against 20 of 20, one-sided $P = 0.77$). Descriptive: the first replicator met its own kind (Hamming distance at most 2) in 57% of its open-phase encounters on the lattice and 11% when mixed. Stage I, lethal tar: a zero byte fetched as an opcode halts the pair for the rest of the encounter (a change to the machine’s semantics, not only to the fate of its by-product), at $L = 16$, ten worlds, seeds 4001–4010, with the culture and partner tests under the same rule; the heritable fraction of random cells rose from 0 at 20,000 steps to 0.41 at 30,000 and 0.88 at 50,000 (medians over worlds). *Heritable material* is dated by the first sampled step at which at least 10% of the 16 random cells of the census are heritable (post hoc, `timing_heritable.py`): a median of 40,000 steps under lethal tar against 1,250 under benign tar (Stage G, $L = 16$), a 32-fold delay, against Kaplan–Meier medians of 38,000 and 450 steps for the first clone, `t_rep`.

BFF. cubff default semantics verified from source: two 64-byte tapes concatenated, pointer and heads at byte 0, ten instructions, every other byte a no-op, unmatched brackets halt, 2^{13} steps per encounter, 2^{17} programs, random ordered pairs each epoch, mutation 2^{-12} per byte per epoch. Switches: `ip_wrap` (pointer wraps modulo 128); `literal` (byte 0x50 = `P a b`: write `a` at `head1 - 1` and `b` at `head1 - 2`, then `head1 - = 2`, skip the operands; designed so that its one-byte tiling writes itself, the analogue of the Z80’s load–push word); `nohalt` (unmatched brackets are no-ops); `halt_p` (an unmatched bracket halts with probability `p`, decided by a hash of the pair, the step and the seed); `lit_rep` (the literal instruction writes `r` copies of its word per execution). Our GPU kernel reproduces cubff’s validation replicator and matches a CPU reference on randomised tests. 24 seeds each for the published and wrapping variants, 12 for each literal variant, 12 for the published rule with no-op brackets (run after review; pre-registered: every first replicator loop-bearing, met 7 of 7; a transition in at least 9 of 12, not met at 7; pointer-closed first replicators in every world, not met in two, one intermediate (pointer enters in 23% of encounters) and one open), 6 for each of $p = 0.01, 0.03, 0.1$ and 0.3 under wrap + literal push, and 6 for each of $r = 2$ and 3 under the literal push without a wrapping pointer; events every 64 epochs; high-order entropy with brotli quality 2 as in ref. 16. Dial predictions, registered before the run: D1-1, the open population persists (share $\geq 10\%$ at 16,384 epochs) in at least 4 of 6 soups at $p = 0.01$ (met, 6 of 6) and in at most 1 of 6 at $p = 0.3$ (not met: the share falls below 1% at 832–896 epochs in 6 of 6 and recovers to 0.10–0.11 in 5 of 6, with 0.31–0.44 of random tapes heritable); D1-2, the epoch of collapse falls monotonically with p within a factor of three of $1/p$ (not met: no collapse at $0 < p \leq 0.1$, 896 at $p = 0.3$, 192 at $p = 1$); D1-3, no closure by control flow at any p (met); D2-1, inflow at most 1 bit at $r \geq 2$ (met: 0.00 bits in 12 of 12 soups at $r = 2$ and 3 , against 8.00 in 12 of 12 at $r = 1$); D2-2, the first replicator still the loop-free, pointer-open tiling (met, 12 of 12); D2-3, collapse slower or absent at $r \geq 2$ (met: absent, with an all- P share at epoch 1,024 of 22–31% at $r = 2$ and of 3.6–7.3% in 5 of 6 soups and 0 in one at $r = 3$, against 0–0.24% at $r = 1$, and every random tape heritable at 16,384 epochs).

Searches. Two-byte census: all 65,536 words tiled to 16 bytes, culture test (32 partners) and partner test (256), plus the deterministic fixed-point test against the zero partner. BFF straight-line search: 16,447,860 programs (prefix ≤ 3 instructions + tiling unit of period 2–5 with a copy instruction), 2 random partners, wrapping pointer, alphabet prefilter then exact similarity, full culture test for survivors. Closed-replicator search: all 1,309,528 periodic tilings of $\{P, [,], \text{no-op}\}$ up to period 10 under the literal, wrapping, no-halt rules, three partners each.

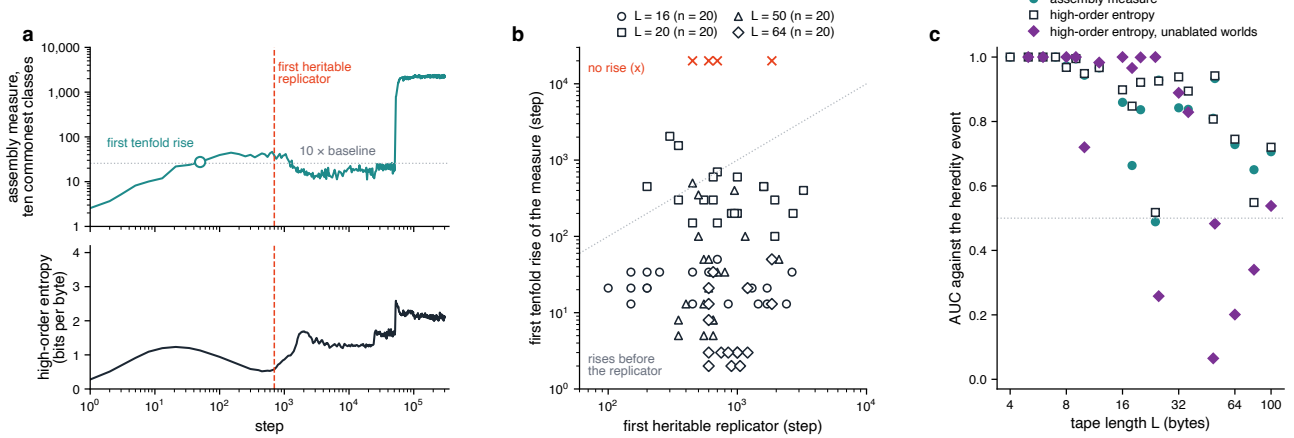
Pre-registration and stages. Extended Data Table 4 lists every run of the numbered stages. Stages D, F, G, H and I, the BFF cells and every stage added in revision (K, L, M, the mutational scan, the pair invasion, the traced execution and the dials) were pre-registered with predictions and kill criteria before launch; every amendment is dated in the change log. Misses are reported in the text: the heritable-fraction thresholds of G1(c) failed at $L = 50$; the syntactic loop criterion of G2 omitted hardware loops (6 of 20 and 3 of 20 syntactic against 19 of 20 and 8 of 20 behavioural); the expectation of 15 of 20 closures at $L = 32$ was not met (12); Stage H’s H1 and H2 were killed (Controls); Stage I’s I2 (the open phase cut short, “open, then extinct” in at least 5 of 10 worlds) was not met (0 of 10) and I3 (closure rarer under lethal tar) was killed (10 of 10 worlds closed); Q1 (in worlds whose modal tape is closed, at least 90% of heritable cells confined) was not met (105 of 109 worlds); the mechanism first claimed for an open BFF replicator under a wrapping pointer was falsified by the harness test before any soup ran. In the second revision the registered tests were serial allele retention, the marker check, population sampling, the constructed-closer invasions, the harness conventions, the aligned invasions, first-closure timing, the census by period, ref. 22’s criterion, the copy-offset switch with its mutational flows, closure under random registers and its replication, the parent $\times$ environment matrix, the matched pair at $L = 32$ under lethal tar, and the lethality dial (REVISION_PREREG R2–R4, E, E-mech, C4a-rep, PE, E-L32L, DZ). Their misses: S1 (the pusher’s lineages stop copying, so its alleles reach first-generation copies only), V1 (the jump spreads at $L = 50$ but confinement does not), P1 (a closed self-writer of period three exists), F1 (ref. 22’s criterion scores pushers and block copiers alike), E1 and E2 (no mutation-bias equilibrium), EM1 (isolated mutational flows do not explain the environment dependence) and C4a (6 of 20 random-register worlds closed, against at most 5 predicted), PE2 (0.36 of the pusher’s lineages survived among partners

drawn from a closed world), the monotone part of DZ2 (closure is slowest at intermediate lethality) and the endpoint count of DZ3 (6 of 10 transmitter-majority worlds at $p = 1$, against at least 7); the motif endpoints of A1, E and C4b failed on definition, since the motifs mutate while their descendants hold the soup, and functional classes of random cells are reported instead. Exploratory stages: A (630 runs, $L = 16$: the unablated soup and six ablations $\times$ budgets {32, 128, 512} $\times$ mutation {1/4, 1/16, 1/64}, ten seeds, 300,000 steps), B (640: the unablated soup and three ablations, block-copy, stack-writes and no-copy, $\times$ budgets {128, 512} $\times$ eight lengths (4, 9, 25, 36, 49, 64, 81 and 100) $\times$ ten seeds), C (490: the unablated soup and 15 ablations at $L = 16$, seeds 101–110, 450 runs, the all-ld, rmw-only and no-copy arms run to one million steps; and the unablated soup and the stack-writes ablation at $L = 36$ and 100, 40 runs), E (1,230: the size axis under four control arms, budget and mutation sweeps at $L = 100$). Stages A and B stopped a run early once quasispecies occupancy reached 50% (112 of 630 and 144 of 640 runs stopped early); later stages have no early stop. Confirmatory stages: D (280: the $L = 9$ reversal, seeds 1001–1020), F (210: padded pair memories), G (80: closure, seeds 2001–2020, $L = 16$ and 50 for 300,000 steps, $L = 20$ and 64 for one million), K, L, M as above.

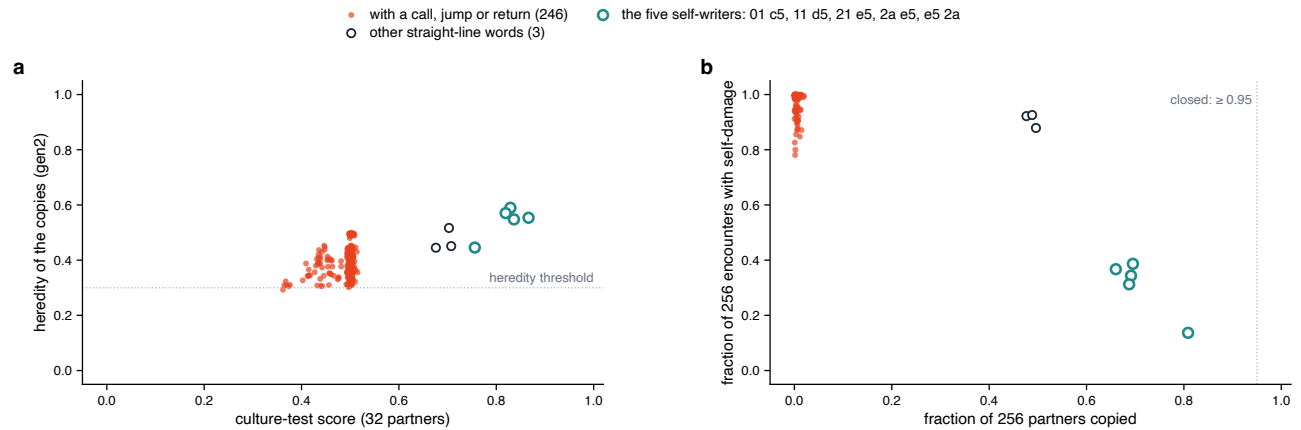
Data availability. Every condition file, raw run summary and sample record, every generated table from which a number in the text or a figure is drawn, and the pre-registration documents with their dated change log are in the project repository, to be archived with a DOI on acceptance. Supplementary Video 1 shows the world of Fig. 2.

Code availability. The simulator (a Z80 emulator in WGSF with the prefix-aware ablation hook, the traced executor and the BFF kernel with its CPU reference), the assays, the analysis scripts and the figure scripts are in the same repository; each figure is produced from the generated tables by one script.

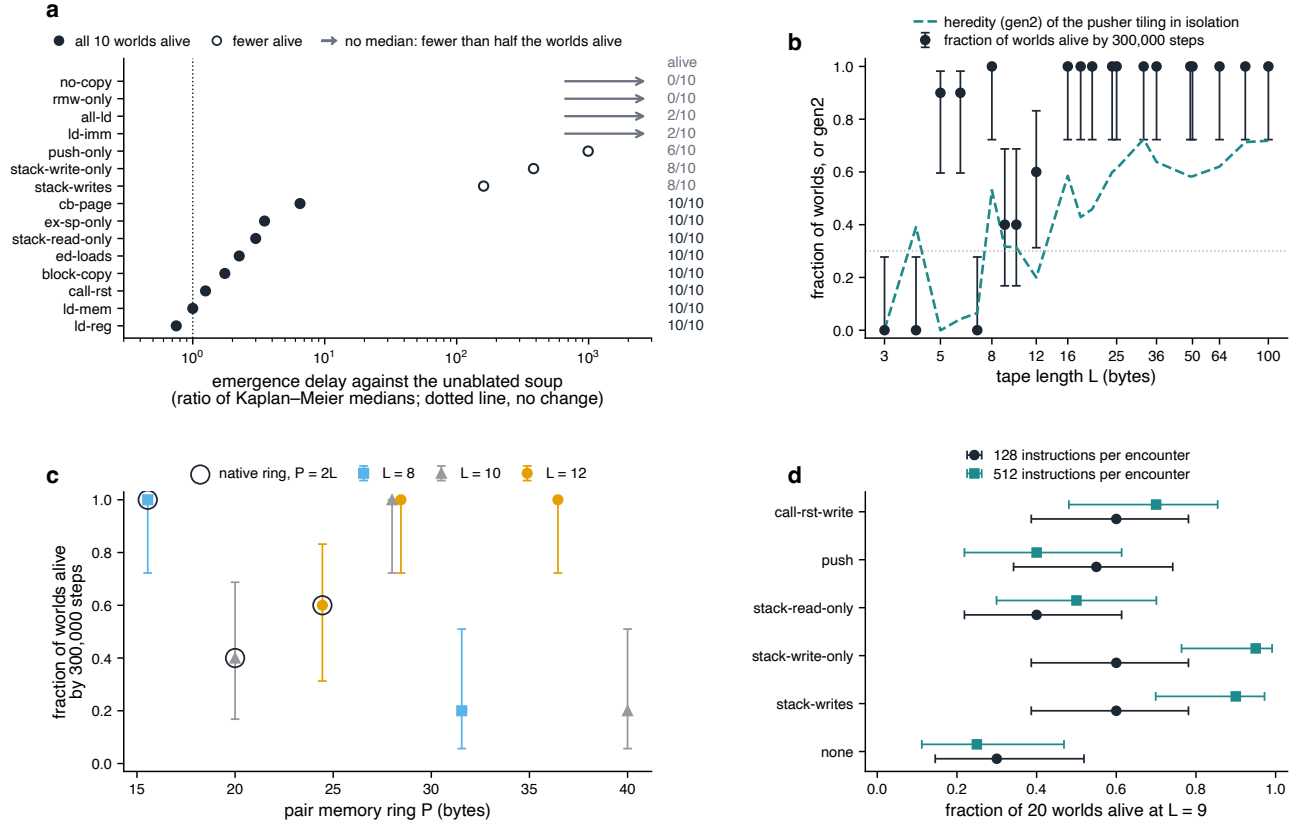
Extended Data



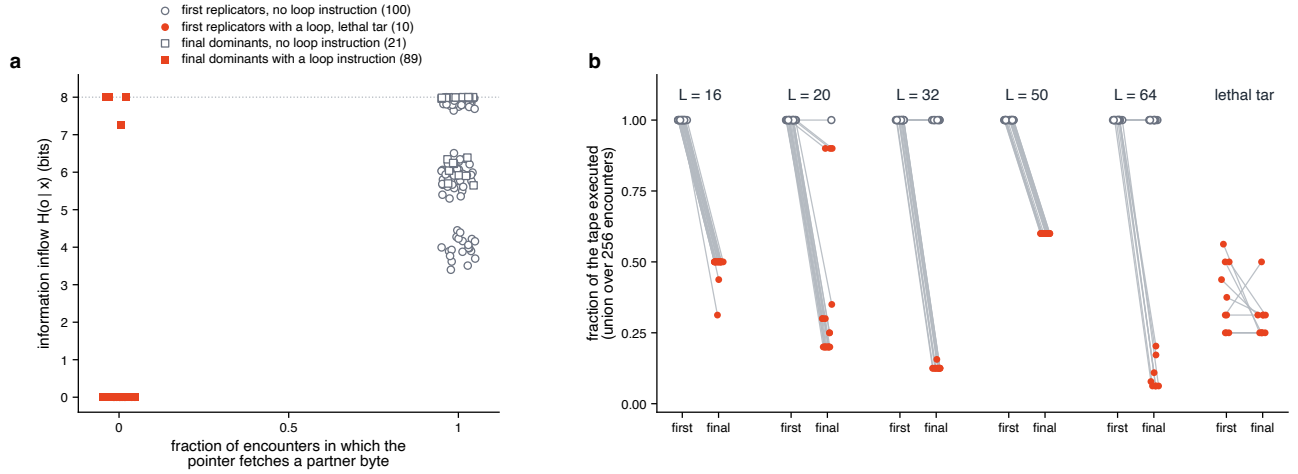
Extended Data Fig. 1 | Two detectors of life against the culture test. **a**, The assembly measure over the ten most common classes (a Re-Pair upper bound on the assembly index in Sharma et al.'s formula; top) and the high-order entropy of ref. 16 (bottom) against step for one world ($L = 16$, seed 2001): the measure rises tenfold above its random-soup baseline at step 50 (circle), 650 steps before the first heritable replicator (dashed line), while the soup is filling with tar. **b**, Step of the measure's first tenfold rise against the step of the first heritable replicator for the 80 Stage G worlds (dotted line, equality; crosses at the top edge, worlds without a rise). **c**, Area under the ROC curve against the heredity event by tape length: the assembly measure (filled circles) and the high-order entropy (open squares) over the four control arms of the size-axis stage at their snapshot steps, and the high-order entropy in unablated worlds alone over every recorded sample (diamonds), near chance or inverted at six of the eight lengths from 25 to 100 bytes; dotted line, chance.



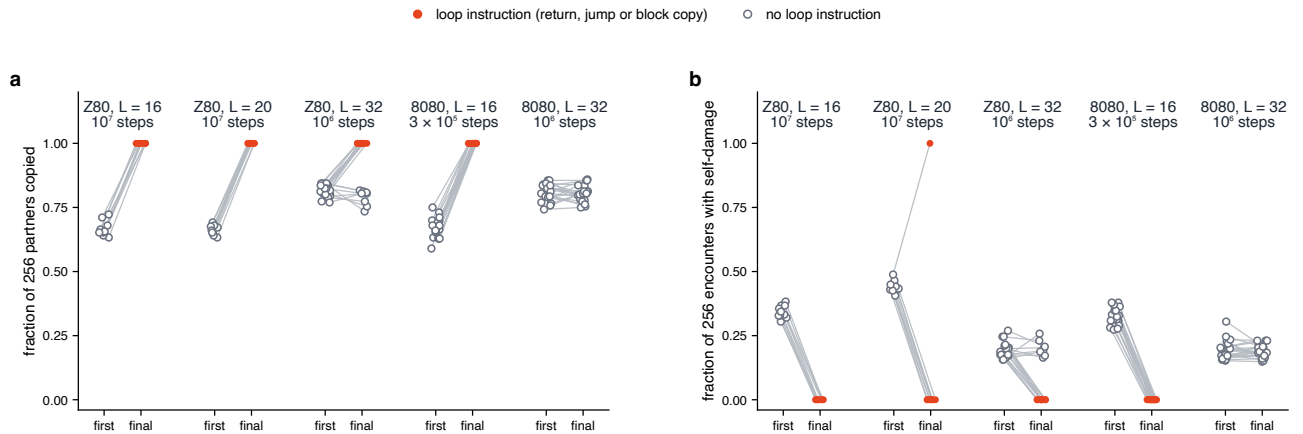
Extended Data Fig. 2 | Every two-byte word. All 65,536 two-byte words were tiled to 16 bytes and tested; shown are the 254 that pass the heredity threshold ($gen2 \geq 0.3$). **a**, The heredity of the copies (gen2) against the culture-test score over 32 random partners. The 246 words that contain a call, jump or return (vermillion) pass by smearing return addresses into the partner: gen2 is a normalised gain in best-shift similarity to the original tape, so smeared return addresses raise the partners' similarity to the word without making copies, and heredity is read together with the copies of the partner test (b). The five load-push words (teal) are exactly the words, besides the trivial $00\ 00$, that write themselves into an all-zero partner (Proposition 3); the three other straight-line words (black) are their push-first rotations. **b**, The partner test over 256 random partners: the fraction of encounters in which the word loses a quarter or more of its own bytes against the fraction of partners that become a $\geq 75\%$ copy; dotted line, the 0.95 that defines a closed copier. No word reaches it, and none of the 246 produces a single 75% copy. Coincident points are spread by a small one-sided jitter.



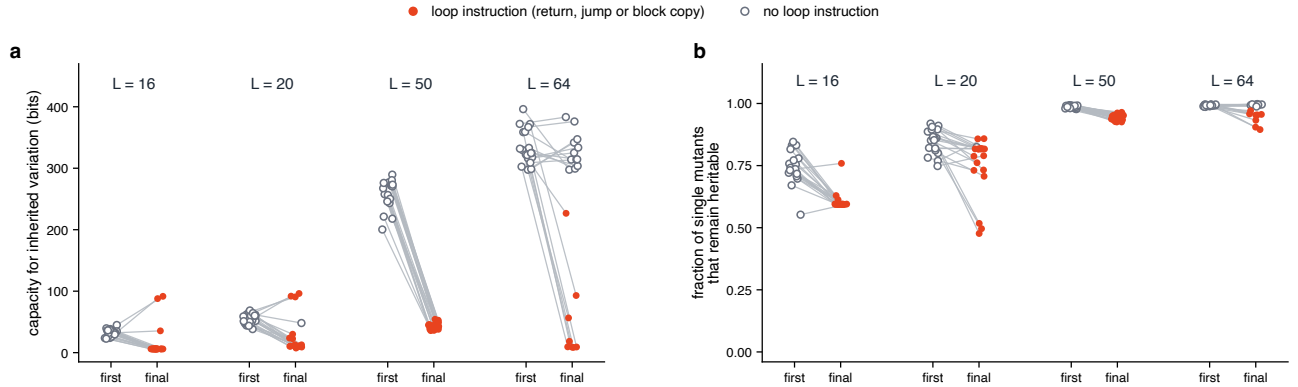
Extended Data Fig. 3 | What the instruction set must provide. **a**, Delay of emergence relative to the unablated soup, as the ratio of Kaplan–Meier medians (128 instructions per encounter, mutation 1/16, seeds 101–110), for each of the 15 ablations of Stage C; filled, all ten worlds alive; open, fewer; arrows, no ratio because fewer than half the worlds produced a replicator; right margin, worlds alive by 300,000 steps, or by 1,000,000 steps for all-ld, rmw-only and no-copy, which were run and counted to one million steps. Among the arms that keep the immediate loads and PUSH, the largest delays are 6.5-fold without the CB page and 3.5-fold without EX (SP),HL (seed-paired sign tests, $P = 0.75$ and 0.18). **b**, Fraction of worlds alive by 300,000 steps against tape length (Wilson 95% intervals; ten worlds per length) and the heredity (gen2) of the pusher tiling tested in isolation at each length (dashed; dotted line, the 0.3 threshold); the two diverge below 16 bytes, where block copiers come first and the stack and instruction pointers collide on the short ring. **c**, Fraction of worlds alive against the ring length P of the pair memory at L = 8, 10 and 12 (Stage F; circled, the native ring P = 2L; Wilson 95% intervals): the outcome depends on the ring as well as on the organism, a 28-byte ring raising it to 10 of 10 at L = 10 and 12 and a 32-byte ring lowering it to 2 of 10 at L = 8. Of the rings shown only P = 16 and 32 are aligned (a ring P is aligned if P divides 65,536, so that the 16-bit program counter and stack pointer fold onto it consistently; Methods). **d**, Fraction of 20 worlds alive at L = 9 by ablation at 128 and 512 instructions per encounter (Stage D): removing the stack writers makes life more likely, not less.



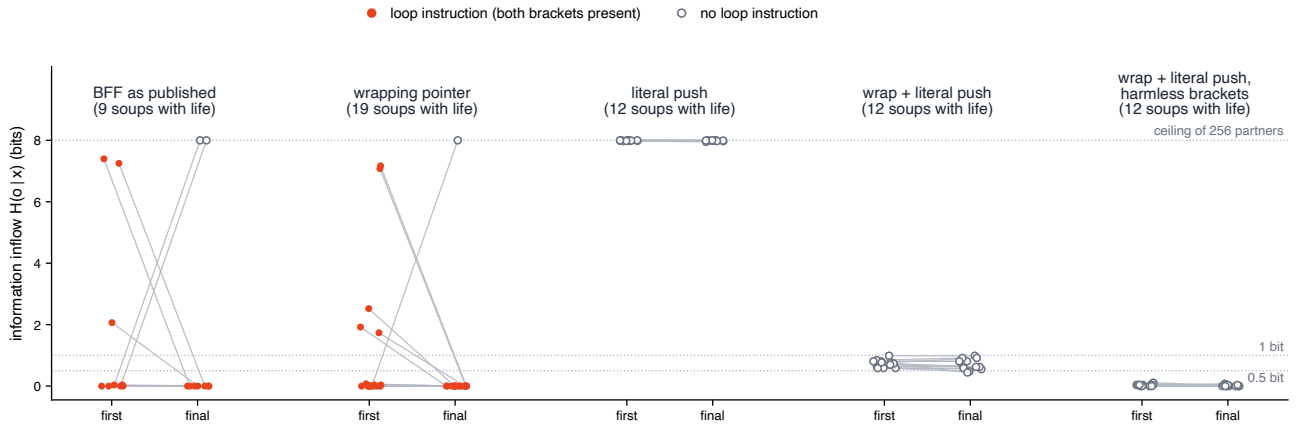
Extended Data Fig. 4 | Pointer confinement, information inflow and the executed part of the tape. Every first replicator (circles) and final dominant (squares) of Stages G, K and I, 110 of each, run against 256 seeded random partners on a traced copy of the executor (filled vermillion, loop instruction; open, none). **a**, Fraction of encounters in which the pointer fetches a partner byte (jittered) against the information the offspring carry about the partner, $H(o|x)$; dotted line, the 8-bit ceiling of 256 partners. The 100 first replicators born under benign tar enter the partner in every encounter and carry 3.4–8 bits; the ten born under lethal tar and all 89 loop-bearing final dominants never enter it, and 85 of the 89 carry no inflow, the other four keeping one to three partner bytes (7–8 bits); the 21 final dominants without a loop instruction are open pushers. **b**, Fraction of the tape fetched as instruction stream in any of the 256 encounters, first replicator against final dominant per world (grey lines join the same world). The pusher executes its whole tape; the closers execute part of it, and the evolved ones regenerate the rest from it (Fig. 4a,c).



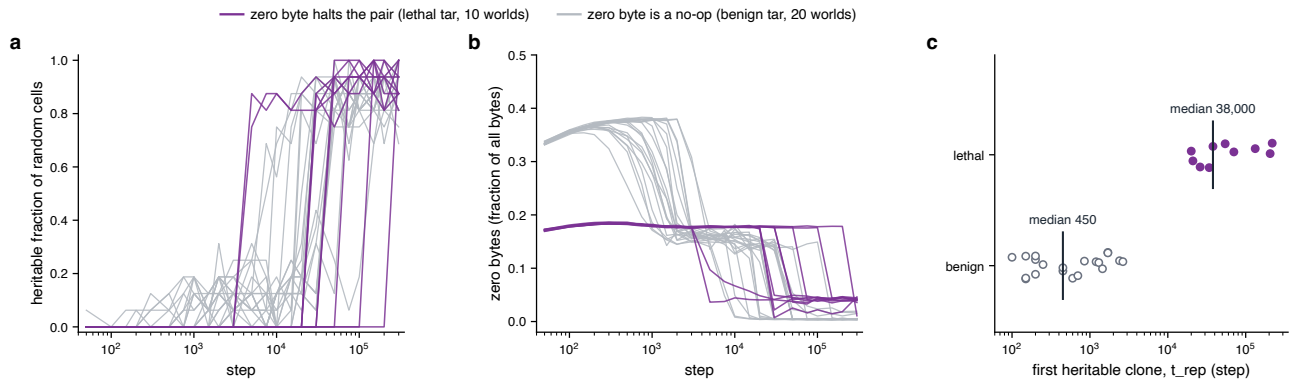
Extended Data Fig. 5 | Closure across lengths, horizons and instruction sets. First heritable replicator (first) and final dominant (final) of every world of the stages added in revision, partner test over 256 random partners (grey lines join the same world; filled, loop instruction). **a**, Fraction of partners copied. **b**, Fraction of encounters with self-damage. Z80 at the aligned length $L = 32$ (Stage K, one million steps): 12 of 20 worlds close, all by block copy. Z80 at $L = 16$ and 20 for ten million steps (Stage L): 20 of 20 close. The 8080 subset (Stage M): at $L = 16$ all 20 worlds close on the return design ad e3 21 e3 21 c0 ad c0; at $L = 32$ none of 20 does within a million steps, although the subset can express a closer that, seeded, takes over (Methods; Extended Data Fig. 9b,c). One $L = 20$ final dominant (seed 6003, a block copier at 0.3% of the soup) damages itself in every encounter while copying every partner.



Extended Data Fig. 6 | Single-mutant scan of first and final dominants. Single-byte mutational scan of the first heritable replicator (first) and the final dominant (final) of every Stage G world, grouped by tape length (grey lines join the same world; filled, loop instruction; open, none); transmission is scored in first-generation copies; serial transfers are in Fig. 4a–c. **a**, Single-mutant variation score, $\sum \log_2(1 + 255 \times \text{fraction of transmissible values})$ over positions, in bits. **b**, Robustness: fraction of single mutants that remain heritable ($\text{gen2} \geq 0.3$). All 255 alternative bytes per position at $L = 16$ and 20, 32 seeded values at $L = 50$ and 64; 32 partners per generation.



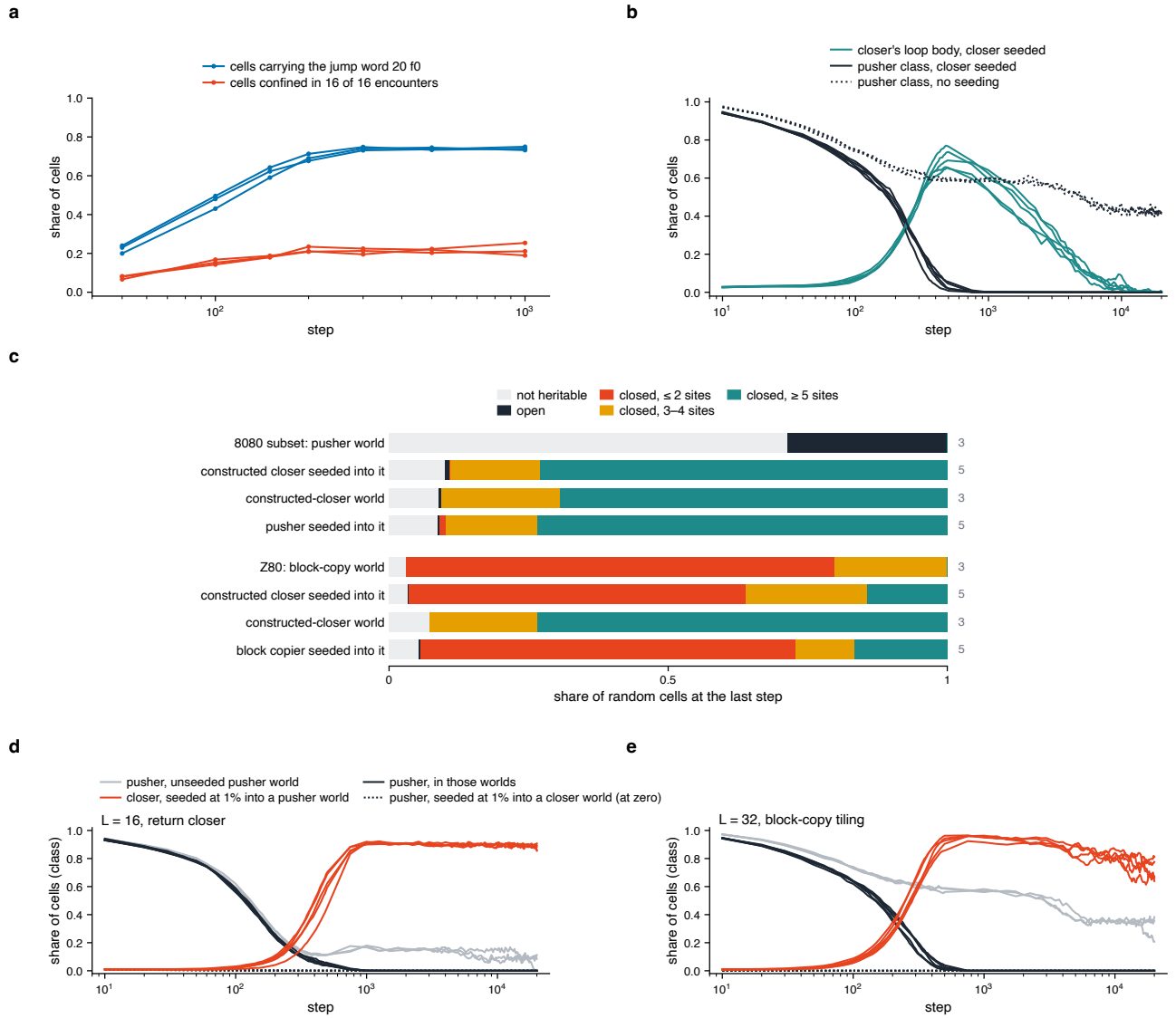
Extended Data Fig. 7 | Information inflow in the second machine. $H(o | x)$ over 256 random partners for the first replicator and the final dominant of every life-producing BFF soup, by variant (grey lines join the same soup; filled, both brackets present, a loop), with the 8-bit ceiling of 256 partners and the pre-registered 1-bit and 0.5-bit thresholds. Without a literal channel (as published, and with a wrapping pointer) every first replicator carries a loop and most carry no inflow; the literal push without a wrapping pointer sits at the ceiling; under a wrapping pointer the literal tiling floods the ring and carries less than 1 bit, with no-op brackets at most 0.11, although its pointer enters the partner in every encounter. The loop-free finals at the 8-bit ceiling are the literal pushers without a wrapping pointer, open copiers (11 of 12 copy 89–94% of partners, the twelfth none) that lie outside Theorem 1, which concerns published BFF programs whose pointer stays in their own cells, and three one-symbol fills (two as published, one with a wrapping pointer) that had displaced the replicator and copy 0.00–0.02 of partners.



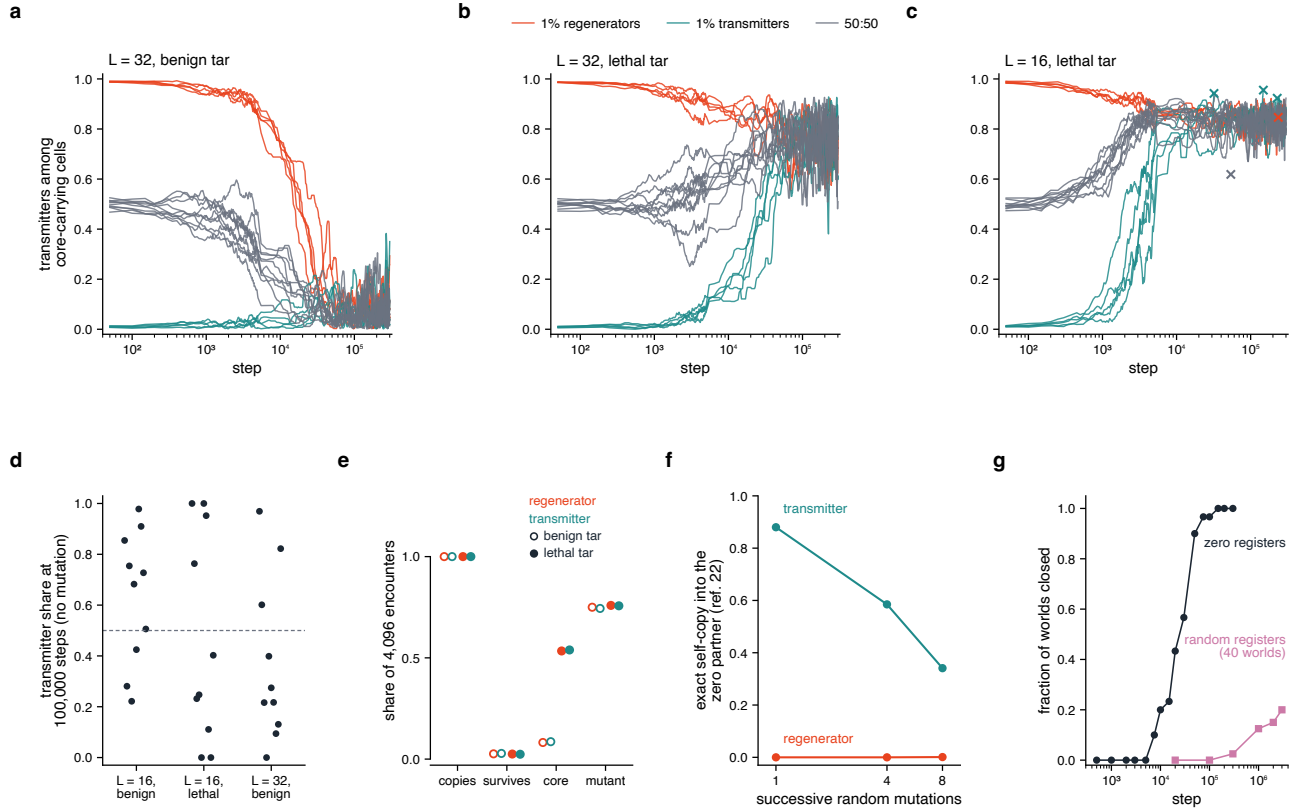
Extended Data Fig. 8 | Lethal tar in the first machine. Ten pre-registered worlds at $L = 16$ in which a zero byte fetched as an opcode halts the pair (Stage I, purple) against the twenty benign-tar worlds of Stage G at the same length (grey). **a**, Heritable fraction of 16 random cells against step. **b**, Zero-byte fraction against step. **c**, Step of the first heritable clone, t_{rep} , in each world (jittered vertically), with Kaplan–Meier medians (38,000 against 450). Every lethal-tar first replicator is a block-copy replicator with a hardware loop, contains no zero byte, and copies every partner without self-damage from its first appearance. The first clone can lag the first heritable material: in two lethal-tar worlds 75–88% of random cells are heritable by step 5,000 (a), while the first tape among the three most common to pass the culture test is dated 21,000 and 206,000 steps; dated by heritable material, the delay is 32-fold (40,000 against 1,250 steps).

Extended Data Table 1 | Final dominants of the confirmatory closure stage, by the instruction that closes the cycle. For each tape length, the 20 final dominants of Stage G (the most heritable of the three most common tapes; at $L = 50$ a stage of a lineage whose offspring reopens, Methods) grouped by the control-flow or block-repeat instruction they carry, with the most common tape of each class (first eight bytes) and the number of worlds that end on it byte for byte, and the medians of the partner test over 256 random partners.

L	instruction that closes the cycle	worlds	most common final tape in the class (worlds)	partners copied	self-damage
16	conditional return (RET NZ)	17	ad e3 21 e3 21 c0 ad c0 ... (17)	1.00	0.00
16	block copy (LDIR)	2	1e 50 f9 c3 8e 4e f6 b5 ... (1)	1.00	0.00
16	absolute jump (JP nn)	1	50 5e c3 eb 12 ee 8a 51 ... (1)	1.00	0.00
20	block copy (LDIR)	13	1e 04 ed b0 1e 04 ed b0 ... (4)	1.00	0.00
20	relative jump (DJNZ)	3	21 e5 21 e5 21 4e 10 e5 ... (2)	1.00	0.00
20	block copy (LDDR)	2	1e 9c 10 f6 6b c2 03 33 ... (1)	1.00	0.00
20	none: the open pusher	1	21 e5 21 e5 21 e5 21 e5 ... (1)	0.69	0.36
20	relative jump (JR)	1	21 e5 21 e5 21 fd 20 e5 ... (1)	1.00	0.00
50	relative jump (JR)	20	01 c5 01 c5 01 c5 01 c5 ... (14)	1.00	0.00
64	none: the open pusher	12	11 d5 11 d5 11 d5 11 d5 ... (5)	0.78	0.06
64	block copy (LDIR)	5	84 5e ed b0 84 5e ed b0 ... (3)	1.00	0.00
64	block copy (LDDR)	2	b8 4b 2c cf ef ec ae 83 ... (1)	1.00	0.00
64	absolute jump (JP (HL))	1	d5 d5 11 73 e9 d5 11 d5 ... (1)	1.00	0.00



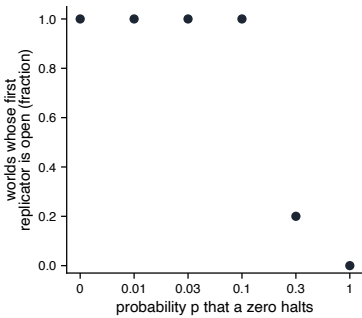
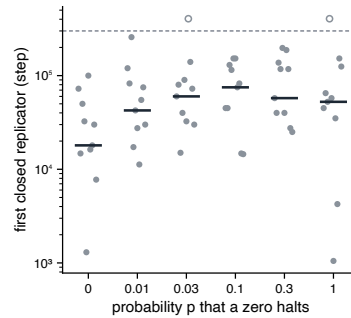
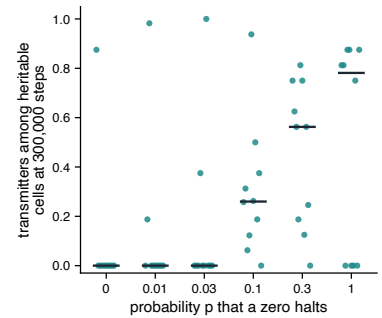
Extended Data Fig. 9 | Five invasion tests. **a**, The $L = 50$ marker against confinement: the closed form with three jumps seeded at 1% into a world of its ancestor (three soups): share of cells carrying the jump word 20 f0 (blue) and share of 512 random cells whose pointer stays out of the partner in 16 of 16 encounters (vermilion), from step 50 (both start at 0.01 at seeding). The jump reaches about 0.74 of the cells; the confined share rises to about 0.2 and levels off. **b**, The constructed 8080 closer (20-byte core and a 12-byte random payload) seeded at 1% into a world of the 8080 pusher at $L = 32$ under the 8080 subset (five soups): share of cells carrying the closer's loop body 2b 46 2b 4e c5 15 c2 (teal; defined after the first run) and share within Hamming distance 8 of the pusher (black, solid), with unseeded pusher worlds (dotted, three soups). The exact loop body then drifts by mutation, so its share falls while the closer's descendants hold the soup (c). **c**, Classes of 64 random cells at the last step (8080 subset, 20,000 steps; Z80, 50,000), mean over soups (numbers at right; classes as in Fig. 4d; the split of closed cells by transmissible sites rests on up to 16 scanned heritable cells per soup). The 8080 pusher world stays open; the constructed closer takes it over, and its descendants carry bytes they never execute; under the full Z80 the evolved block-copy tiling 04 5e ed b0 $\times 8$ holds its world against the constructed closer and, seeded at 1%, displaces it. **d, e**, Invasions at aligned lengths, under the Stage G dynamics at $L = 16$ (d) and the Stage K dynamics at $L = 32$ (e): the evolved return closer ad e3 21 e3 21 c0 ad c0 $\times 2$ and the evolved block-copy tiling 04 5e ed b0 $\times 8$ seeded at 1% into a world filled with the pusher (closer class, vermilion; pusher class in the same soups, black; five soups each), the pusher seeded at 1% into a world filled with the closer (dotted, five soups; at zero within 140–270 steps at $L = 16$ and 20–30 at $L = 32$), and unseeded pusher worlds (grey, three each; at $L = 16$ one of them closed by itself). A class is the set of cells within Hamming distance $L/4$ of the design at any cyclic shift. At 20,000 steps every sampled heritable cell of the closer-seeded soups is confined. Colours as in Fig. 4: vermilion, closers that regenerate (d, e); teal, the transmitting constructed closer (b).



Extended Data Fig. 10 | The copy-offset switch and closure without supplied registers. a–c, A regenerator R and a transmitter T with the same core, $XX\ 5e\ ed\ b0$, differing only in the offset byte and in what follows the core (L = 32: R = $04\ 5e\ ed\ b0 \times 8$, T = $a0\ 5e\ ed\ b0 + 28$ random bytes; L = 16: R = $44\ 5e\ ed\ b0 \times 4$, T = $b0\ 5e\ ed\ b0 + 12$ random bytes, drawn for every cell), with the standard mutation: transmitters among the cells carrying the core against step, from 1% regenerators (vermillion), 1% transmitters (teal) or a 50:50 mixture (grey); one line per world (five, five and ten); after step 5,000 each line is a running median of nine samples, and samples with fewer than 1,000 core-carrying cells are then omitted; x, the last sample before a world lost the core. At L = 32 every start converges to a low transmitter share with benign tar (a) and to a high one with lethal tar (b); at L = 16 with lethal tar (c) to a high one, although the core was later lost in 7 of 20 worlds (1 of 5 from 1% regenerators, 3 of 5 from 1% transmitters, 3 of 10 mixtures). At L = 16 with benign tar and mutation the core fell below 1,000 cells in all 20 worlds, and that condition is not shown; 7 of the 20 were above 1,000 again at 300,000 steps, with transmitter shares of 0.14–0.63. **d,** Without mutation the share drifts: transmitter share at 100,000 steps in each of ten 50:50 worlds per condition (dashed, the start). **e,** Encounter components of the pristine pair at L = 16 (4,096 encounters each; partners and executors drawn from the final soups of the mixed worlds of each tar, under benign tar soups in which the core had collapsed): exact copy into a partner (copies), exact survival as the partner of a soup executor (survives), core bytes 1–3 intact as a partner (core), and exact copying by a random single mutant (mutant); hollow markers, benign tar; filled, lethal tar. **f,** Post hoc, ref. 22's robustness criterion for the L = 32 pair (exact self-copy into the all-zero partner, 512 instructions, after k successive random mutations; 1,000 trials; median over 20 transmitter tails): the transmitter scores high because it keeps mutations, the regenerator near zero because it erases them. **g,** Fraction of worlds at L = 16 in which most heritable cells among 64 random cells are confined, at the recorded snapshots, with zero registers (black: Stages G and L, 30 worlds; 20 at 7,500 steps) and with every register drawn at random before each encounter (pink: 40 worlds, seeds 8101–8120 and 8201–8220).

a

	random partners, benign	random partners, lethal	closed benign world	lethal world	open phase (step 5,000)	random registers
pusher	○	○	○	○	○	○
return closer	●	○	●	○	●	○
block-copy R	●	●	●	●	●	○
block-copy T	●	●	●	●	●	○
lethal-tar closer 4009	●	●	●	●	●	○
self-initialising closer	●	●	●	●	●	●

b**c****d**

Extended Data Fig. 11 | Parent, environment and lethality. **a**, Six parents at $L = 16$ copied through eight serial transfers in six environments, in the order of the columns (every single-byte mutant in 16 lineages, 256 unmutated lineages): benign or lethal rule with uniform random partners; partners drawn from the closed final soup of Stage G world 2001 (benign rule), from the lethal final soup of Stage I world 4001 (lethal rule) or from the open phase of world 2001 at step 5,000 (benign rule); and random registers with random partners. Filled, unmutated lineages alive after eight transfers (1.00 in every filled cell); hollow, lost (at most 0.04 alive, except the pusher among partners from the closed world, 0.36, drawn hollow because fewer than half survive); number, transmissible sites after eight transfers. Colours: open pusher, black; regenerators, vermilion; transmitters, teal. The mode belongs to the parent; the environment decides which parents survive. **b–d**, The lethality dial: a zero byte fetched as an opcode halts the pair with probability p , ten worlds per p at $L = 16$ (the same seeds at every p), 300,000 steps. **b**, Fraction of worlds with a replicator whose first replicator is open (its pointer enters the partner in at least half of 16 traced encounters); at $p = 1$ one world (seed 9004) has no heritable common tape and the fraction is of 9. **c**, Step of the first closed replicator (a top-3 tape holding at least 0.5% of cells, heritable and confined in 16 of 16 encounters, the registered endpoint) per world; hollow circles, not closed by 300,000 steps (dashed); bars, Kaplan–Meier medians (the fifth of ten ordered times). The endpoint misses closed populations of transmitters, whose genomes all differ so that no tape holds 0.5% of cells: both hollow worlds ($p = 0.03$, seed 9007; $p = 1$, seed 9004) are closed and transmitter-dominated at 300,000 steps (d), which biases the timing test against closure at high p . **d**, Share of heritable random cells that are transmitters (confined, at least five transmissible sites) at 300,000 steps per world; bars, medians (the mean of the fifth and sixth of ten).

Extended Data Table 2 | Instruction-set ablations. Each ablation replaces the listed instruction families or mnemonics by no-ops through a prefix-aware fetch hook; the count is of opcodes on the base, CB and ED pages (indexed DD/FD forms follow their base opcode); a leading minus removes a family from the set. i8080 is the 8080 subset of Stage M.

ablation	opcodes made no-ops	patterns (instruction families or mnemonics)
none	0	(none)
block-copy	4	block-copy
stack-writes	46	stack, ex, call-ret, rst
stack-write-only	22	PUSH, EX (SP),HL, CALL, RST N
stack-read-only	24	POP, RET, EX DE,HL, EXX, EX AF,AF'
push-only	8	stack
push	4	PUSH
ex-sp-only	1	EX (SP),HL
call-rst	34	call-ret, rst
call-rst-write	17	CALL, RST N
ld-mem	31	ld8-mem, ld16-mem
ld-imm	11	ld8-imm, ld16-imm
ld-reg	54	ld8, ld-special
all-ld	96	ld8, ld8-imm, ld8-mem, ld16-imm, ld16-mem, ld-special
ed-loads	8	8 ED-page opcodes
cb-page	256	rotate, rotate-mem, bit, bit-set, bit-set-mem, -RLD, -RRD
no-copy	112	ld8, ld8-imm, ld8-mem, ld16-imm, ld16-mem, ld-special, stack, ex, block-copy
rmw-only	46	writes-mem, -incdec-mem, -rotate-mem, -bit-set-mem
i8080	342	bit, bit-set, bit-set-mem, rotate, rotate-mem, EX AF,AF', DJNZ, JR, EXX, 78 ED-page opcodes

Extended Data Table 3 | Uncertainty across worlds and threshold sensitivity. Medians over worlds with 95% bootstrap intervals (10,000 resamples of worlds; Kaplan–Meier medians of t_{rep} with 2,000 resamples, censored at the horizon). Counts with Wilson 95% intervals: first replicator a load–push word, benign tar, L 16–64: 170 of 170 (0.98–1.00); worlds closed at the population level, Stage G, L = 16: 20 of 20 (0.84–1.00); L = 20: 19 of 20 (0.76–0.99); L = 32 (Stage K): 12 of 20 (0.39–0.78); L = 50: 0 of 20 (0.00–0.16); L = 64: 8 of 20 (0.22–0.61); 8080 subset, L = 32: 0 of 20 (0.00–0.16). Threshold grid: the first replicator of every benign-tar world of 16–64 bytes (Stages G, H, K, L and M) re-dated from the recorded runs under gen2 thresholds of 0.2 to 0.6 (the paper uses 0.3) and informative-partner thresholds of 0.65 to 0.85 (the paper uses 0.75; it changes nothing): gen2 threshold 0.2: load–push first in 170 of 170 worlds with a replicator (0.98–1.00); open first in 170; gen2 threshold 0.3: load–push first in 170 of 170 worlds with a replicator (0.98–1.00); open first in 170; gen2 threshold 0.4: load–push first in 170 of 170 worlds with a replicator (0.98–1.00); open first in 170; gen2 threshold 0.5: load–push first in 155 of 169 worlds with a replicator (0.87–0.95); open first in 155; gen2 threshold 0.6: load–push first in 107 of 164 worlds with a replicator (0.58–0.72); open first in 99. At thresholds of 0.5 and above the pusher’s own gen2 (0.46–0.62) falls near the threshold, so later closers are dated first in some worlds.

quantity	condition	median	95% bootstrap interval	worlds
first replicator: share of 256 random partners copied	Z80, L = 16	0.68	0.65–0.70	20
first replicator: share of encounters with self-damage	Z80, L = 16	0.33	0.31–0.35	20
first replicator: gen2	Z80, L = 16	0.58	0.58–0.58	20
first replicator: share of partners holding an exact copy	Z80, L = 16	0.58	0.55–0.59	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 16	450	200–1,200	20
first: share of single mutants heritable	Z80, L = 16	0.74	0.72–0.78	20
final: share of single mutants heritable	Z80, L = 16	0.59	0.59–0.60	20
first replicator: share of 256 random partners copied	Z80, L = 20	0.66	0.64–0.67	20
first replicator: share of encounters with self-damage	Z80, L = 20	0.43	0.42–0.45	20
first replicator: gen2	Z80, L = 20	0.46	0.44–0.46	20
first replicator: share of partners holding an exact copy	Z80, L = 20	0.32	0.31–0.33	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 20	700	550–1,000	20
first: share of single mutants heritable	Z80, L = 20	0.86	0.82–0.88	20
final: share of single mutants heritable	Z80, L = 20	0.82	0.75–0.82	20
first replicator: share of 256 random partners copied	Z80, L = 50	0.68	0.66–0.70	20
first replicator: share of encounters with self-damage	Z80, L = 50	0.11	0.10–0.12	20
first replicator: gen2	Z80, L = 50	0.58	0.58–0.63	20
first replicator: share of partners holding an exact copy	Z80, L = 50	0.05	0.04–0.05	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 50	600	500–650	20
first: share of single mutants heritable	Z80, L = 50	0.99	0.98–0.99	20
final: share of single mutants heritable	Z80, L = 50	0.95	0.94–0.95	20
first replicator: share of 256 random partners copied	Z80, L = 64	0.73	0.72–0.75	20
first replicator: share of encounters with self-damage	Z80, L = 64	0.06	0.06–0.07	20
first replicator: gen2	Z80, L = 64	0.62	0.62–0.62	20
first replicator: share of partners holding an exact copy	Z80, L = 64	0.00	0.00–0.00	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 64	750	600–1,000	20
first: share of single mutants heritable	Z80, L = 64	0.99	0.99–0.99	20
final: share of single mutants heritable	Z80, L = 64	0.99	0.96–0.99	20
first heritable clone t_{rep} , Kaplan–Meier median (steps)	Z80, L = 16, lethal tar	38,000	21,000–132,500	10
last snapshot: share of heritable random cells confined	Stage G, L = 16	1.00	1.00–1.00	20
last snapshot: share of random cells heritable	Stage G, L = 16	0.86	0.85–0.88	20
last snapshot: share of heritable random cells confined	Stage G, L = 20	1.00	0.99–1.00	20
last snapshot: share of random cells heritable	Stage G, L = 20	0.94	0.92–0.95	20
last snapshot: share of heritable random cells confined	Stage G, L = 50	0.29	0.27–0.30	20
last snapshot: share of random cells heritable	Stage G, L = 50	0.55	0.54–0.57	20
last snapshot: share of heritable random cells confined	Stage G, L = 64	0.18	0.09–1.00	20
last snapshot: share of random cells heritable	Stage G, L = 64	0.28	0.23–0.97	20
last snapshot: share of heritable random cells confined	Stage I, L = 16	1.00	1.00–1.00	10
last snapshot: share of random cells heritable	Stage I, L = 16	0.93	0.92–0.95	10
last snapshot: share of heritable random cells confined	Stage K, L = 32	1.00	0.00–1.00	20
last snapshot: share of random cells heritable	Stage K, L = 32	0.94	0.35–0.96	20
last snapshot: share of heritable random cells confined	Stage L, L = 16	1.00	1.00–1.00	10
last snapshot: share of random cells heritable	Stage L, L = 16	0.86	0.85–0.88	10
last snapshot: share of heritable random cells confined	Stage L, L = 20	1.00	1.00–1.00	10
last snapshot: share of random cells heritable	Stage L, L = 20	0.95	0.94–0.96	10
last snapshot: share of heritable random cells confined	Stage M, L = 16	1.00	1.00–1.00	20
last snapshot: share of random cells heritable	Stage M, L = 16	0.86	0.85–0.88	20
last snapshot: share of heritable random cells confined	Stage M, L = 32	0.00	0.00–0.00	20
last snapshot: share of random cells heritable	Stage M, L = 32	0.49	0.36–0.54	20

Extended Data Table 4 | Every Z80 soup run of the numbered stages. From the run summaries: arms (their number, or the arm itself: `none`, the unablated soup; `mixed`, well-mixed pairing; `lethal`, lethal tar; `i8080`, the 8080-like subset; the ablations are listed in Extended Data Table 2), tape lengths L, budgets (instructions per encounter), background mutations per step, seeds, horizons (k, thousand steps; M, million) and the runs stopped early by the occupancy rule of Stages A and B; 3,660 runs in all. The revision-2 tests (harness conventions, invasions, the copy-offset switch, the lethality dial) are listed in Methods.

stage	runs	arms	L	budgets	mutations / step	seeds	horizon	early stops
A	630	7	16	32, 128, 512	128, 512, 2048	1–10	300k	112
B	640	4	4, 9, 25, 36, 49, 64, 81, 100	128, 512	512	1–10	300k	144
C	490	16	16, 36, 100	32, 128, 512	128, 512, 2048	101–110	300k, 1M	0
D	280	6	9	32, 128, 512	128, 512	1001–1020	300k	0
E	1230	4	3–100 (21)	32–2048 (13)	32–4096 (20)	1–1020 (31)	300k	0
F	210	none, stack-write-only	8, 10, 12, 16, 36	128	512	101–110	300k	0
G	80	none	16, 20, 50, 64	128	512	2001–2020	300k, 1M	0
H	10	mixed	16	128	512	3001–3010	300k	0
I	10	lethal	16	128	512	4001–4010	300k	0
K	20	none	32	128	512	5001–5020	1M	0
L	20	none	16, 20	128	512	6001–6010	10M	0
M	40	i8080	16, 32	128	512	7001–7020	300k, 1M	0

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Supplementary Information: theorems and proofs

What kind of statement each item is: a **Theorem** has a pen-and-paper proof and uses no simulation; a **Proposition** is an exact statement decided by finite computation, with the computation described; a **Model** is a result about a stylised model with stated assumptions, not about the simulator. Empirical premises are marked as such, and every number quoted is from the generated tables behind the main text.

Notation. An encounter executes a program held in a shared memory. The organism A occupies addresses 0 to $L - 1$ and the partner B the next L addresses; in the Z80 soups the $2L$ addresses form a ring, in BFF the 128 addresses do not wrap. Execution starts at address 0 with a fixed initial machine state (registers zero; the Z80 stack pointer at the last byte of B) and lasts at most T instruction executions ($T = 128$ in the Z80 soups, 2^{13} in BFF). A *control transfer* is a jump, call, return, repeat-in-place or halt; a *straight-line program* executes none, so that under default flow its pointer advances past each executed instruction. The *no-op context* is the partner all of whose bytes are no-ops (zero in the Z80, where byte 0 is NOP; zero in BFF, where every byte outside the ten instruction characters is a no-op). A tape x is a *self-writer* if its encounter with the no-op context ends with B holding a copy of x (up to a cyclic shift, as in the culture test), and a *full self-replicator in context e* if the encounter (x, e) ends with B holding a copy of x . It is *closed* if its pointer executes no address of B in any context and *open* if it does in some context. A value is *prescribed* if it is determined by x alone, the same for every partner.

Theorem 1 (the cost of a copy in BFF). *Setting.* BFF as published: a 128-byte pair; the instruction pointer p and the two data heads h_0 and h_1 start at address 0; p advances by one after every instruction unless a bracket jumps; $< >$ and $\{ \}$ move h_0 or h_1 by one cell; $+ -$ change the byte under h_0 by one; $.$ copies the byte under h_0 into the cell under h_1 and $,$ the byte under h_1 into the cell under h_0 ; every other byte is a no-op; execution ends when p leaves the pair, on an unmatched bracket, or after 2^{13} steps. *Statement.* Let A be the organism’s 64-byte tape, at addresses 0–63, fix a cyclic shift s , and let B be a partner whose byte differs in every cell from the copy it is to receive (A rotated by s). Any execution of the pair that leaves B holding that copy runs at least 128 instructions: at least 64 that change a cell and at least 64 head moves. If p stays among A’s 64 addresses throughout, some address is therefore executed at least twice. *Proof.* Every cell of B must change, and an instruction changes at most one cell ($+$, $-$ and $,$ the cell under h_0 , $.$ the cell under h_1), so at least 64 changing instructions run, and each cell of B is under a head when it changes. Both heads start at address 0, which lies in A, and each move changes one head’s position by one, so after v moves the heads have occupied at most v positions besides 0; occupying all 64 cells of B needs $v \geq 64$. Moves change no cell, so the two sets of instructions are disjoint, and at least 128 instructions run. An execution of at least 128 instructions at 64 addresses executes some address twice. ■ *Corollary 1.1.* A BFF program that copies its tape, at some shift, into a partner differing in every cell from that copy

while its pointer stays in its own cells executes a loop in that encounter. *Scope*. The count is per encounter: it needs neither the same instruction sequence in every encounter nor a no-op partner, and it says nothing about a program whose pointer enters the partner, which then fetches the partner's bytes. An earlier version stated two further parts, a bound uniform over partners for programs whose pointer stays out of B and the claim that no straight-line execution writes a copy of a tape with more than 32 non-zero bytes into a zero partner. Both proofs were wrong, the first because brackets test the byte under h0 and , copies B's bytes into A, so that a confined program's control flow can depend on B, the second because bytes written into B ahead of the pointer are executed when the pointer arrives; both parts are withdrawn. That every observed BFF first replicator carries a loop (35 of 35 under the published rule, the wrapping pointer and harmless brackets) is an observation explained, not forced, by the count; whether the loop also keeps the pointer out of B is a separate, measured fact (22 of the 28 first replicators under the published rule and the wrapping pointer never entered the partner in 256 encounters).

Theorem 2 (closure requires a cycle). *Hypothesis (H1), default flow*. In the absence of an executed control transfer the pointer advances past each instruction, so that a straight-line program runs from the organism's region into the partner's. *Statement*. Under (H1), a closed full self-replicator of L bytes executes some address at least twice, provided either (i) a copy of it costs more than L instruction executions, or (ii) execution cannot end inside the organism's region before a budget $T > L$ is spent. *Proof*. A closed execution visits only the L addresses of A. If no address repeats, it comprises at most L instruction executions. Under (i) these cannot complete the copy; under (ii) the budget forces a further execution, which can only be at an address already executed. ■ *The machines*. BFF satisfies (i) in every encounter with a partner that differs in every cell from the copy it is to receive: by Theorem 1 the copy costs at least 128 executions, more than the organism's 64 addresses. The Z80 soups with benign tar satisfy (ii): the budget is 128 executions for tapes of at most 100 bytes, there is no exit instruction, and HALT holds the processor at its own address, which we count as a one-state cycle. *Where the provisos fail*. Under lethal tar (Stage I) a zero byte fetched as an opcode ends execution, possibly inside the organism, and in the Stage E arms with budgets of 32 and 64 instructions at $L = 100$ the budget ends before a pass over the tape; there (ii) does not hold and the theorem does not apply. *The bandwidth reading*. Condition (i) holds in any machine whose straight-line code writes fewer bytes than the addresses it consumes, since a pass over L addresses then writes fewer than L bytes. This is the form used in the main text, and it describes the replicators observed: the first replicator writes two bytes per four bytes of code (the executor trace shows eight bytes written when the pointer leaves its sixteen cells), BFF's copy instructions move one byte per instruction and need a head move per cell, and the literal push of modified BFF writes two bytes per three. It is not a machine-wide property of the Z80, in which a chain of pushes writes two bytes per byte of code; for the Z80 the theorem rests on (ii). *Corollary 2.1*. Under (H1) every straight-line program is open: with $T > L$ its pointer reaches the partner's region. A closed self-writer therefore contains an executed control transfer, which may be a repeating instruction such as LDIR, besides the instructions that write its content. For the Z80 the census of Proposition 3 makes this exact at small periods: the self-writers of period two are five straight-line load-push words, all open, and the smallest closed self-writer has period three (b0 5e ed tiled, whose tiling contains the block copy b0 5e ed b0). The corollary does not make closed self-writers rare: at period four they outnumber the open ones.

Proposition 3 (the short self-writers of the Z80; computer-assisted). Tile a word of period k to 16 bytes and execute it for 128 instructions as A against the all-zero partner; it is a self-writer if B ends holding the tiling at some cyclic shift. By exhaustive execution of every word: at period two, of 65,536 words exactly six leave B holding a tiling of the word, the trivial 00 00, which B already holds, and the five load-push words 01 c5, 11 d5, 21 e5 (LD rr, nn ; PUSH rr) and 2a e5, e5 2a (LD HL, (nn) ; PUSH HL); none contains a control transfer, so each runs on into the partner, and every two-byte self-writer of this machine is open. At period three, of 16,777,216 words exactly one is a self-writer, b0 5e ed, which is closed: tiled, it contains the block copy b0 5e ed b0 (OR B ; LD E, (HL) ; LDIR), whose copy offset is the tape length. At period four, of 4,294,967,296 words, 641 are self-writers of primitive period four: 558 closed, every one of them a block copy (455 with LDIR, 103 with LDDR), and 83 open. Closure is tested by tracing the self-writer against the zero partner and 16 random partners (closed if the pointer fetches no partner byte in any). The stochastic census of period two agrees: the same five words copy into 0.66–0.81 of 256 random partners, none into 0.95 or more, and the 246 words that pass the heredity threshold by smearing return addresses produce no 75% copy. *What it shows*. The smallest self-writers are open, and a closed self-writer needs only one period more; but closed self-writers are the majority at periods three and four, the largest the census covers, because a single repeating block move closes a copy. Counting self-writers does not, therefore, explain why the open word comes first.

Proposition 4 (a count on the initial soup). In a soup of N tapes of L uniformly random bytes, the expected number of occurrences of a given k-byte word, at any of the $L - k + 1$ positions of a tape, is $N(L - k + 1) / 256^k$: for $N = 20,000$ and $L = 16$, 4.6 for a given two-byte word, 0.017 for a given three-byte word and 6.1×10^{-5} for a given four-byte word. *What the count does and does not show*. The load-push word works only in phase, as the four bytes 01 c5 01 c5 (LD BC, nn reading c5 01, then PUSH BC), the same length as the shortest evolved closers (1e a4 ed b0, 44 5e ed b0), so a given open and a given closed working unit have the same expected count at step 0, 6.1×10^{-5} . Proposition 3 counts

more closed than open four-byte self-writers. The count therefore gives the open form no head start; the head start is measured (Kaplan–Meier medians of 450 steps to the first replicator and 21,000 to the first closed replicator in Stage G at $L = 16$; 38,000 under lethal tar, where the first replicator is closed). An earlier version compared a two-byte word with a four-byte closer and concluded that the open form is exponentially likelier; that comparison was wrong and is withdrawn. Why the open word is found first, perhaps because partial load–push words write useful bytes from any position along an executed path while a closer works only from the start of execution, is not modelled.

Model 5 (the closure window). *Assumptions.* An open replicator population of $n(t)$ tapes; each tape-step produces a closed functional variant with probability q (a mutation placing the right bytes at the right phase: q is of order $\text{sites} \times \mu \times 256^{-m}$ for an m -byte closing motif); a closed variant, once present, grows faster than the open form and fixes with probability of order one. *Statement.* The expected number of closed variants produced while the open population lasts is $q \int n dt$, and closure occurs with probability $\approx 1 - \exp(-q \int n dt)$; the integral is the window. If the open form’s own copies are viable in the soup it creates, n stays near the carrying capacity, the window grows without bound and closure is a matter of waiting; if the soup becomes lethal to the open form before $q \int n dt$ reaches order one, the open form dies without closing. *What the experiments found.* Z80, benign zeros: closure of the population in 20 of 20 worlds by 300,000 steps at $L = 16$, in 19 of 20 by one million at $L = 20$ and in 8 of 20 at $L = 64$, where the open form is least damaged and the motif must land in the longest tape (smallest q); at $L = 50$ a closed tape arises in 20 of 20 worlds but its offspring reopens, so the model’s assumption that a closed variant breeds true fails there. BFF with the literal push and lethal brackets: under a wrapping pointer the open form held 0.58–0.63 of $N = 2^{17}$ tapes at epoch 64 and $2-3 \times 10^{-4} N$ at epoch 256, an integrated window $\int n dt$ of $8.2-8.7 \times 10^6$ tape-epochs over 16,384 epochs, $5.0-5.5 \times 10^6$ of them before epoch 256; without a wrapping pointer the window was $2.5-4.1 \times 10^5$; 0 of 12 closures in each. BFF with benign brackets: the window is unbounded, yet no closure by control flow in 12 of 12, consistent with $q = 0$, since no closed heritable tiling exists in that language up to period 10 (Methods); the open form persists for 16,384 epochs. Z80 with lethal zeros (Stage I): the open form never establishes, so the window is empty from the start, and the first replicator is closed and late (Kaplan–Meier median 38,000 steps against 450; 10 of 10 worlds). The registered lethality dial (zeros halt with probability p) tested the model’s monotone prediction and refuted it: the first closed replicator comes at Kaplan–Meier medians of 18,000, 42,500, 60,000, 75,000, 57,500 and 52,500 steps at $p = 0, 0.01, 0.03, 0.1, 0.3$ and 1, so a damaged open phase delays closure beyond either extreme, and the window reading holds only for the contrast of the extremes. At the aligned length $L = 32$, 12 of 20 worlds closed by one million steps, and given ten million steps 20 of 20 closed at $L = 16$ and 20. In a world filled with the open form from step 0 at $L = 50$, n at carrying capacity throughout, the jump arose by mutation and spread to half the cells within 5,000–13,250 steps in 3 of 3, though its carriers are mostly not confined. The model’s qualitative prediction, that an open phase feeds closure, is met at $L = 16$: the first closed replicator appears at a Kaplan–Meier median of 22,500 steps under benign tar against 38,000 under lethal tar, where there is no open phase (log-rank $P = 0.007$), although descent of the closers from the open form is not shown. When an unmatched bracket halts with probability p , the open form held 98% of the soup at $p \leq 0.1$ and collapsed at $p = 0.3$ and 1; the registered expectation that the collapse time would scale as $1/p$ was not met, the lethality acting as a threshold between 0.1 and 0.3. When the literal writes two or three copies of its word, so that a copy completes in one pass, no partial copies are made, no lethal tar forms and the open form persists for 16,384 epochs in 12 of 12 soups, with zero inflow from birth. The model does not predict which regime a machine is in; that depends on how the open replicator’s own by-products act on it, which is what the experiments measure.

Theorem 6 (architecture decides the first replicator). *Hypotheses.* (H1) Default flow, as in Theorem 2, with execution ending when the pointer leaves the pair (as in BFF) or continuing around a ring (as in the Z80 soups). (H2) *Copy cost:* in the no-op context, an execution that runs each address at most once cannot write a copy of the tape. (H2) is a hypothesis to be checked per machine. We do not prove it for BFF, where writes can land ahead of the pointer and be executed; a search of 16,447,860 straight-line BFF programs under a wrapping pointer found none that copies a tape of period two or more, only one-symbol fills (Methods). (H3) *Self-consistent literal write:* an instruction that writes its own operand bytes verbatim at a write pointer that starts in, or advances into, the partner’s region, and whose operand can be its own encoding in the phase executed (LD BC, nn ; PUSH BC with $nn = c5\ 01$; P a b with $a = b = P$). *Statement.* (i) Under (H1) and either proviso of Theorem 2, every closed self-replicator contains a cycle. (ii) Under (H1) and (H3), the tiling of the literal instruction is a straight-line program, hence open, that writes its own content at the advancing write pointer; it is a self-writer wherever the encounter lets the written bytes cover the partner’s region, which holds in the Z80 (eight pushes over two passes of the 2L-byte ring; Proposition 3); in BFF with P the tiling is observed to arise first (below), which the theorem does not decide. (iii) Under (H1) and (H2) without (H3), in a machine whose execution ends when the pointer leaves the pair, no straight-line program is a self-writer and every self-replicator contains a cycle: born loop-bearing (whether its pointer also stays out of the partner is a separate, measured property). (iv) Under uniform random initialisation over an alphabet of size σ , the expected number of occurrences of a fixed k -symbol pattern among N regions of length L is $N(L - k + 1) \sigma^{-k}$, so a minimal open pattern shorter than every closed one is present with probability higher by a factor

exponential in the length difference. This concerns the presence of the pattern, not of a replicating tiling, and nucleation is not modelled; in the Z80 the minimal working units, open and closed, are both four bytes long, so (iv) gives the open form no advantage there (Proposition 4). *Proof.* (i) is Theorem 2. (ii) The tiling executes no control transfer, so by (H1) its pointer runs into the partner's region: it is open; its writes are its operands, by (H3) the tiling itself; completion of the copy is the per-machine fact cited. (iii) In the no-op context a straight-line program runs each address at most once before the pointer leaves the pair, and by (H2) such an execution cannot complete a copy; so a self-writer executes some address twice. (iv) is linearity of expectation. ■ *What it decides.* Whether a substrate's first replicator can be open is decided by (H3); whether it is open in practice is observed. The Z80 has a self-consistent literal write (open first, 80 of 80 worlds); BFF has none, since its instructions increment in place or copy from memory (28 of 28 first replicators loop-bearing, and none of the 28 enters the partner in more than 34% of encounters, 22 of them never); BFF with one literal-write instruction added is open first (12 of 12 soups under each pointer rule). The theorem says nothing about how long the open phase lasts (Model 5). *Scope.* (H1) is a property of a von Neumann architecture in which a program's code lies in a shared memory and execution runs on past it: it holds in the Z80 soups, BFF, Core War and Tierra, and not in Avida, whose genomes are circular, so that a straight-line Avida program returns to its own start. The theorem also inherits the provisos of Theorem 2, which fail under lethal tar and in budget arms shorter than the tape. (H2) is a property of the copy instructions and must be checked per machine; Theorem 1 does not imply it (one pass over 128 addresses meets Theorem 1's bound), and we do not prove it for BFF. In the ring Z80 the premise that execution ends when the pointer leaves the pair fails, so (iii) does not apply there: the two-byte self-writer completes its copy on a second pass of the ring. For the Z80 the open-first conclusion rests on observation; the census (Proposition 3) shows that the smallest self-writers are open, but not why they come first. For a chemistry the hypotheses are a reading, not a theorem: (H1) would be a copier that runs off its template into adjacent material by default, (H3) a template whose content specifies its own complement without an interpreter.

Lemma 7 (closure as zero information inflow). Let the offspring of x in context e be $o = F(x, e)$, with e drawn from a context distribution, and define the inflow $\lambda(x) = I(o; e | x)$, the information the offspring carries about the context given the parent. (i) If $\lambda(x) > 0$ then $P(o = x | x) < 1$: a replicator with inflow cannot be perfectly faithful across contexts. (ii) If $P(o = x | x) = 1$ then $\lambda(x) = 0$: perfect fidelity across contexts implies zero inflow. (iii) Fano's inequality bounds the fidelity: $P(o \neq x | x) \geq (H(o | x) - 1) / \log|O|$, where O is the set of possible offspring. *Proof.* If $o = x$ with probability one given x , then o is a function of x alone and $I(o; e | x) = 0$, which is (ii); (i) is its contrapositive; (iii) is Fano's inequality with $H(o | x) \geq I(o; e | x)$. ■ *Measurement.* For a deterministic executor $H(o | x, e) = 0$, so $\lambda(x) = H(o | x)$, the entropy of the offspring over partners; the partner test estimates it as the plug-in entropy over 256 uniformly random partners (ceiling 8 bits). *Reading, with the measured qualification.* The lemma is Krakauer et al.'s individuality, the organism's future determined by its own past rather than the environment's, specialised to one generation and made measurable by intervention. Pointer closure (execution never enters the partner) is neither necessary nor sufficient for $\lambda = 0$: it is not sufficient because offspring positions the organism never writes keep the partner's bytes (7 of the 22 BFF first replicators whose pointer never entered the partner have $H(o | x) > 0$, three of them ≈ 7 bits from a single unwritten byte), and it is not necessary because a copier that overwrites everything it touches has $\lambda = 0$ although its pointer enters the partner (the BFF literal that writes two or three copies of its word, whose offspring carry 0 bits of the partner while its pointer enters the partner in every encounter; Fig. 5d). Inflow comes from partial copying; openness of control flow is the mechanism that makes the Z80 first replicator's copies partial. Measured: the first replicator has $\lambda > 0$ in 80 of 80 worlds (medians 3.9, 6.2, 7.8 and 8.0 bits at $L = 16, 20, 50$ and 64 , the last at the ceiling) and the loop-bearing successor $\lambda = 0$ in 63 of 67 (one offspring string in all 256 contexts), the four exceptions copying all but one to three bytes (Fig. 3c).

Proposition 8 (the copy period). *Setting.* The Z80 soup at tape length $L \leq 64$: the pair memory is a ring of $2L$ bytes, addresses are taken mod $2L$, every register except the stack pointer is zero at the start of an encounter, and an encounter executes 128 instructions, each repetition of $LDIR$ counting as one. Let the organism's tape A begin with $XX\ 5e\ ed\ b0$, where XX is a one-byte instruction that writes no memory and leaves $HL \equiv 0$ and $256 \cdot D \equiv 0 \pmod{2L}$ and BC either 0 or at least 126, as every working value does (Methods), and let $d = XX \bmod 2L$ satisfy $4 \leq d \leq L$ and $d | L$ (hence $d | 2L$). *Statement.* Whatever the partner holds, the encounter leaves both the organism's and the partner's L bytes equal to the d -periodic tiling of A 's first d bytes. Hence a mutation at a position i with $4 \leq i < d$ is carried by the offspring and by all its descendants, whatever the mutant byte; a mutation at a position $i \geq d$ is erased from the offspring and from the parent; and outside the four-byte core the number of transmissible sites is exactly $d - 4$. *Proof.* The first instruction writes nothing; the second, $LD\ E, (HL)$, reads address 0 and sets $E = XX$, so $DE \equiv XX \equiv d \pmod{2L}$; $LDIR$ then writes, at its k -th repetition ($k = 0, 1, \dots$), the current byte at address k into address $d + k \pmod{2L}$, and because BC is 0 (65,536 repetitions) or at least 126 it repeats for all of the 126 executions left. By induction on k , the byte read at address k equals A 's original byte at $k \bmod d$: for $k < d$ the address has not yet been written (the first write to it comes at repetition $2L - d + k > k$), and for $k \geq d$ it was written at repetition $k - d$ with the original byte at $(k - d) \bmod d = k \bmod d$. So every written address a receives the original byte at $a \bmod d$ (for writes that wrap past address $2L - 1$ this uses $d | 2L$), which for $a < d$ is the byte already there.

The writes cover addresses $d, \dots, d + 125$, which include every address from d to $2L - 1$ because $2L - d \leq 124$; addresses below d are unchanged. Hence every address a holds A 's original byte at $a \bmod d$, and because $d \mid L$ the partner's position j , at address $L + j$, holds the byte at $j \bmod d$. Bytes 4 to $d - 1$ are never executed (execution stays at addresses 0–3), so their values do not affect the copy and are copied to every position congruent to them mod d ; the offspring therefore begins with the same core and the same $d - 4$ bytes, and the argument repeats in every generation. Bytes at positions d and beyond are replaced by copies of earlier bytes in both tapes. ■ *Remarks.* The variant `le NN ed b0 (LD E,n ; LDIR)` has offset $d = NN \bmod 2L$ and the same count. Mutations inside the core either preserve its function for a few values (another working value of XX with the same $d - 4$ bytes) or change or break the copy; the single-mutant scan counts a site only if most of its values are carried, so the core contributes none. The law is confirmed in the soups (Fig. 6c): of the 2,343 confined block-copy cells whose offset divides L , 2,289 have exactly $d - 4$ transmissible sites in the single-mutant scan, the median equals $d - 4$ in all 15 (L, d) groups, and the three cells whose offset does not divide L have 0, 3 and 0 sites. A transmitter is the case $d = L$: it copies itself verbatim and carries $L - 4$ sites; every $d < L$ regenerates, rewriting the organism's own body from its first d bytes.